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(54) **SYSTEMS AND METHODS FOR USING VIRTUAL REALITY TO SIMULATE A WORK TASK USING A VOCATIONAL MASK AND BIDIRECTIONAL COMMUNICATION BETWEEN AT LEAST TWO USERS**

(71) Applicant: **BlueForge Alliance**, Bryan, TX (US)

(72) Inventor: **Arnold Kravitz**, St. Petersburg, FL (US)

(73) Assignee: **BlueForge Alliance**, Bryan, TX (US)

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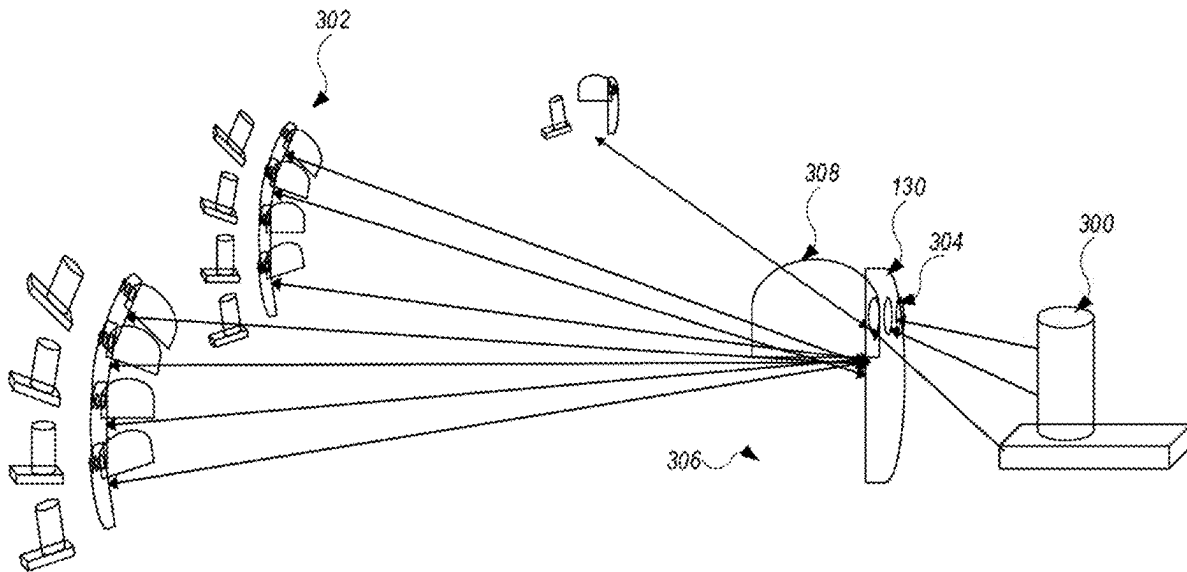
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(57) **ABSTRACT**

Systems and methods for using virtual reality to simulate a work task using a vocational mask and bidirectional communication between at least two users are disclosed. A vocational mask configured to be worn by a first user includes a virtual retinal display, a memory device storing instructions and a processing device coupled to the memory device and the virtual retinal display. Executing the instructions causes the vocational mask to share a virtual reality session with a computing device associated with a second user. The vocational mask may conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master. The vocational mask may receive and present, data in various formats from the second user. The data comprises directions for performing a task by the first user under direction from the second user.



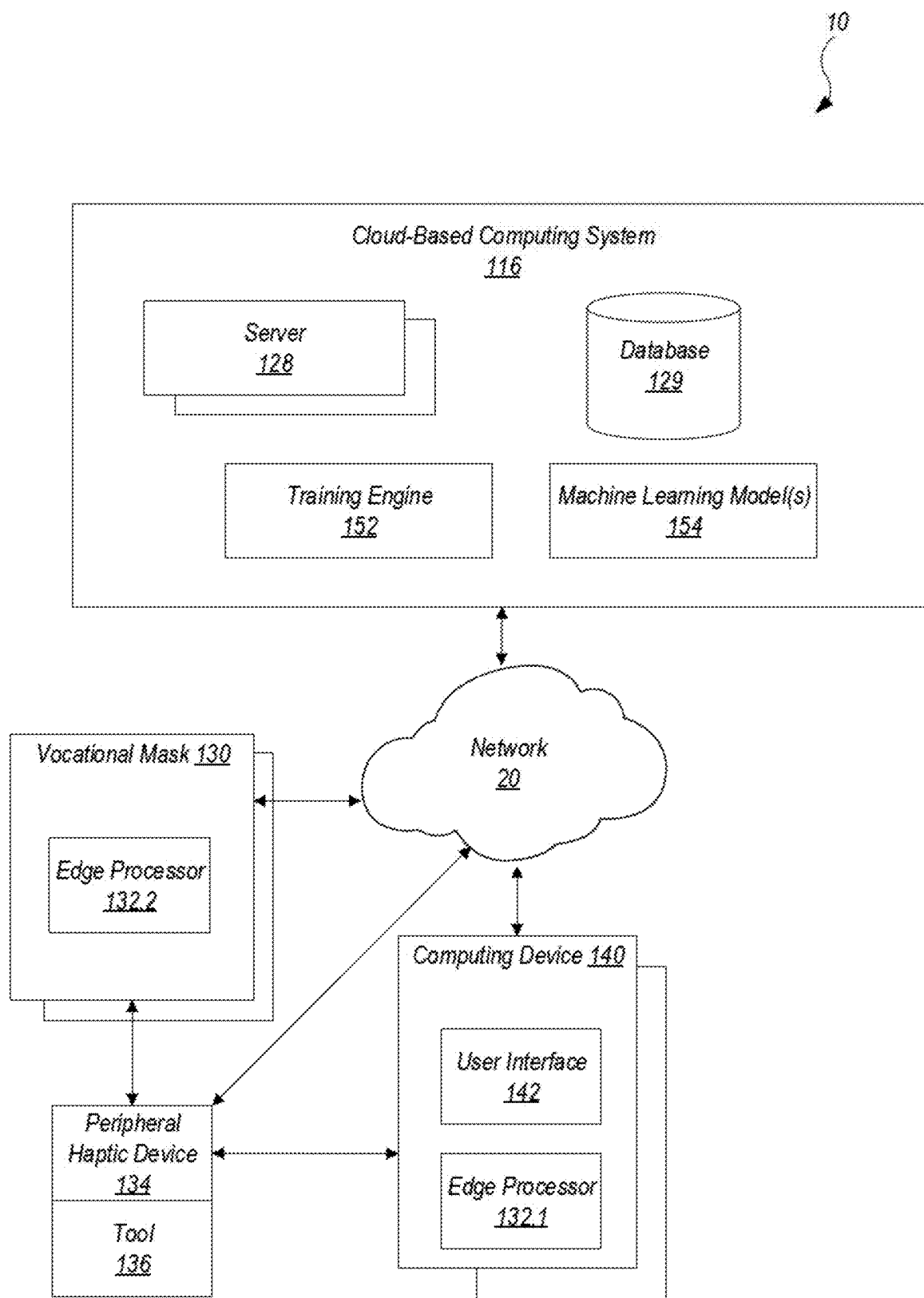


FIG. 1

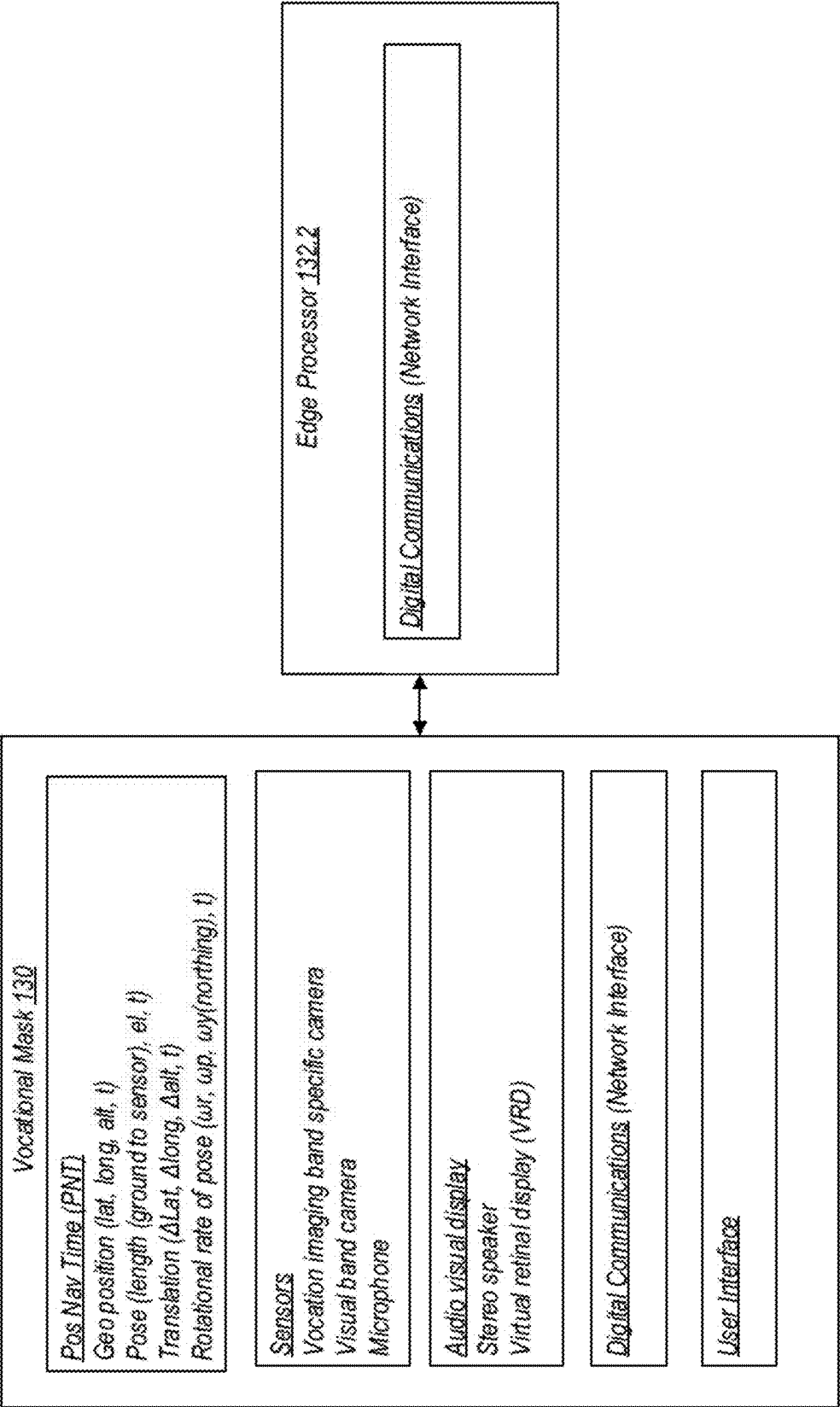


FIG. 2

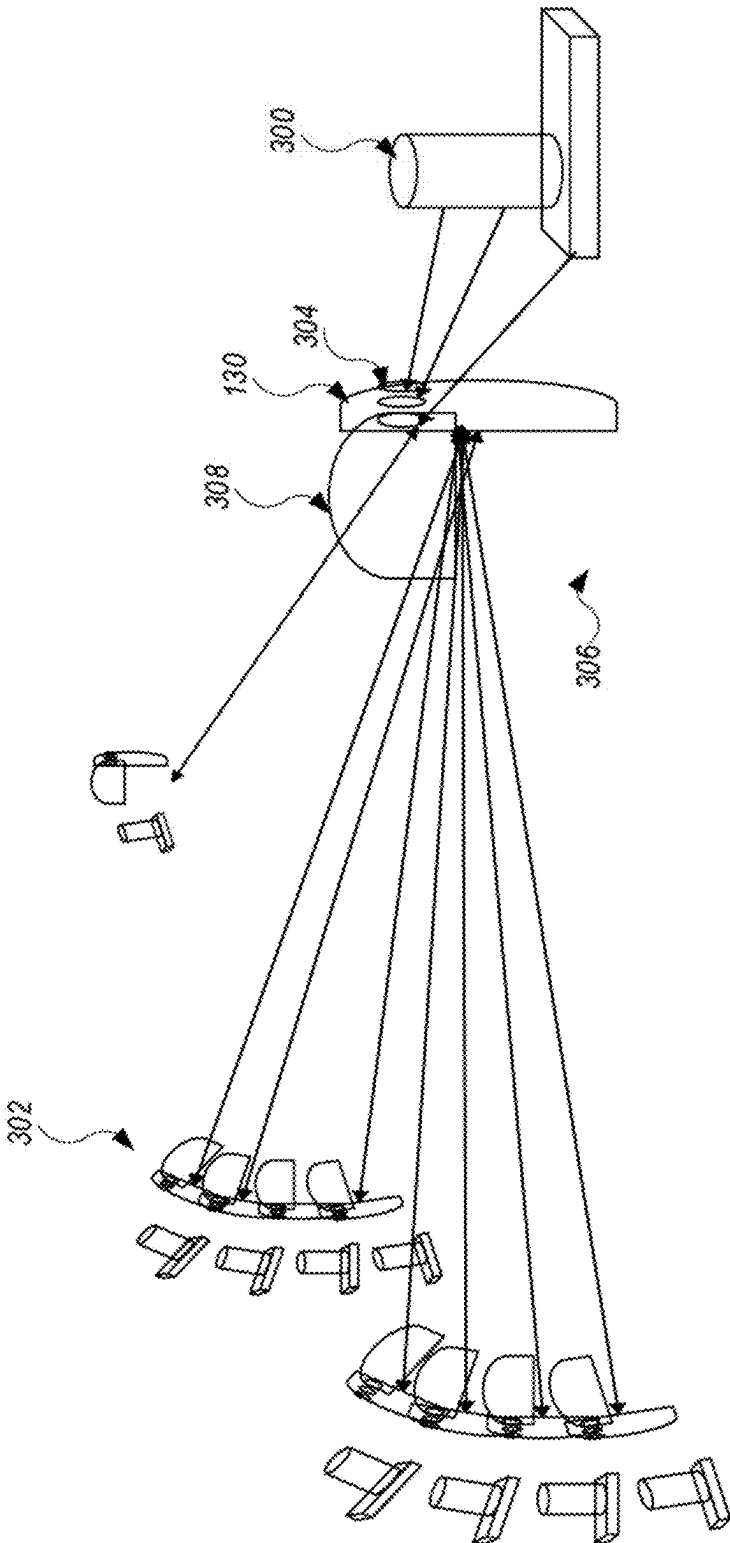


FIG. 3

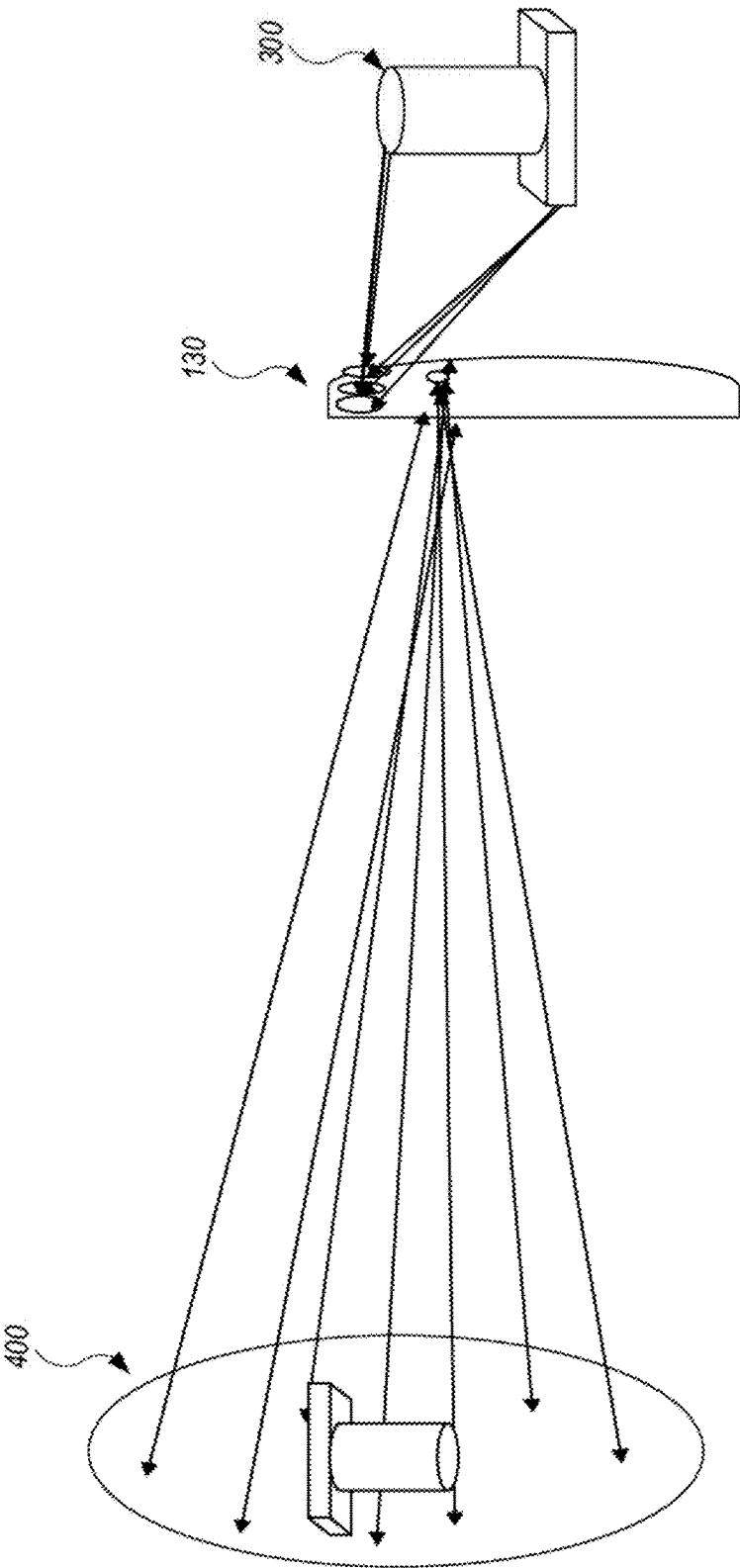


FIG. 4

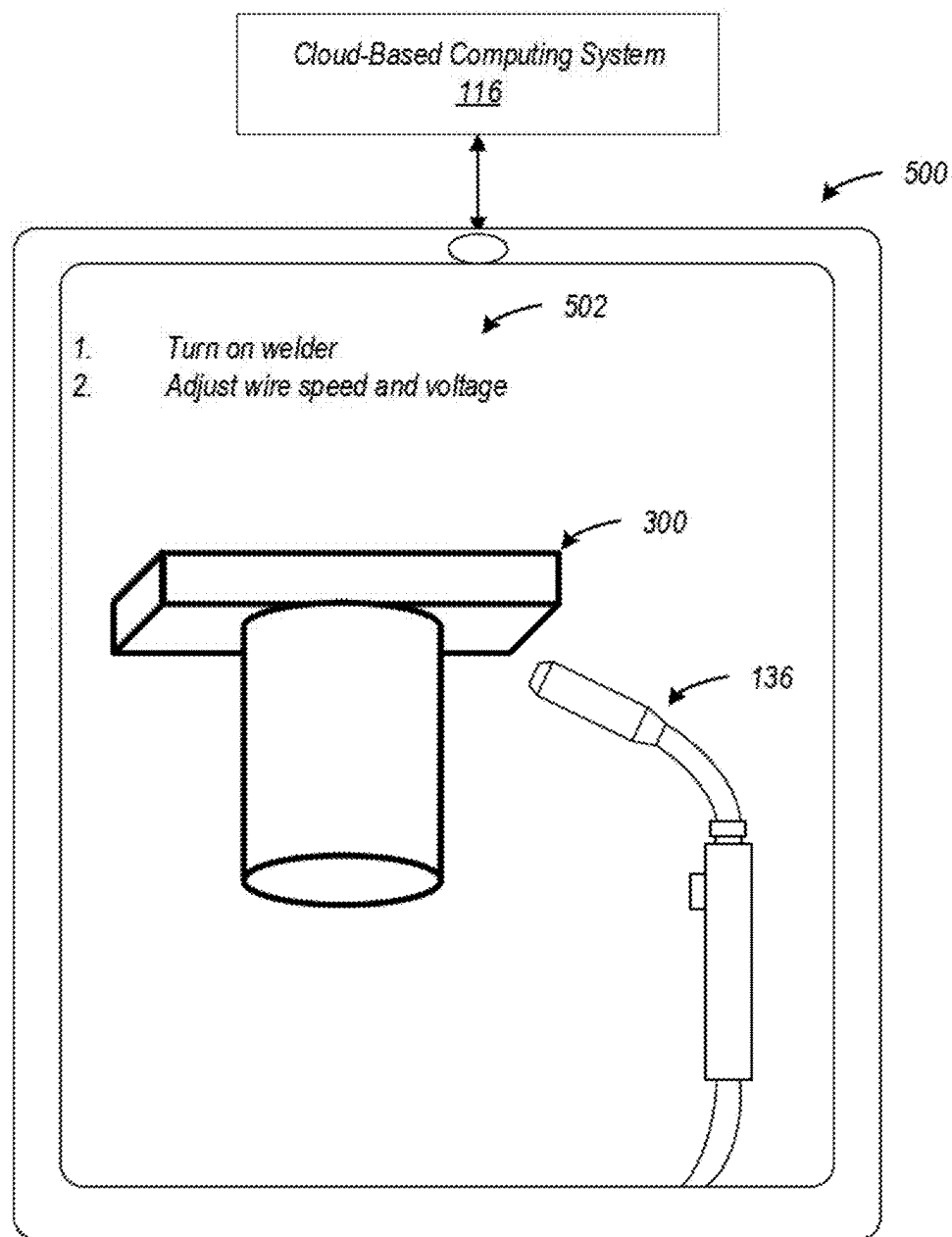


FIG. 5

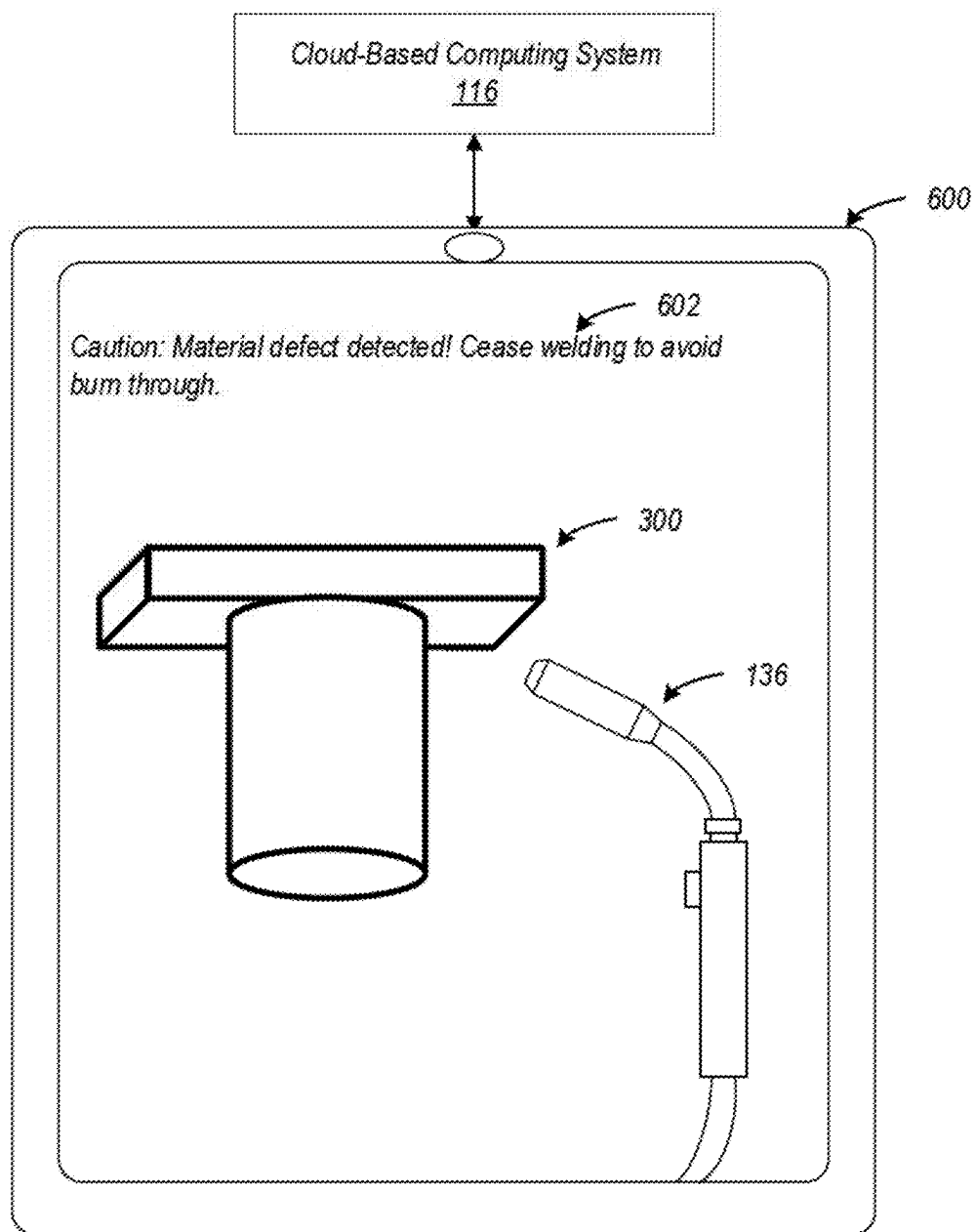


FIG. 6

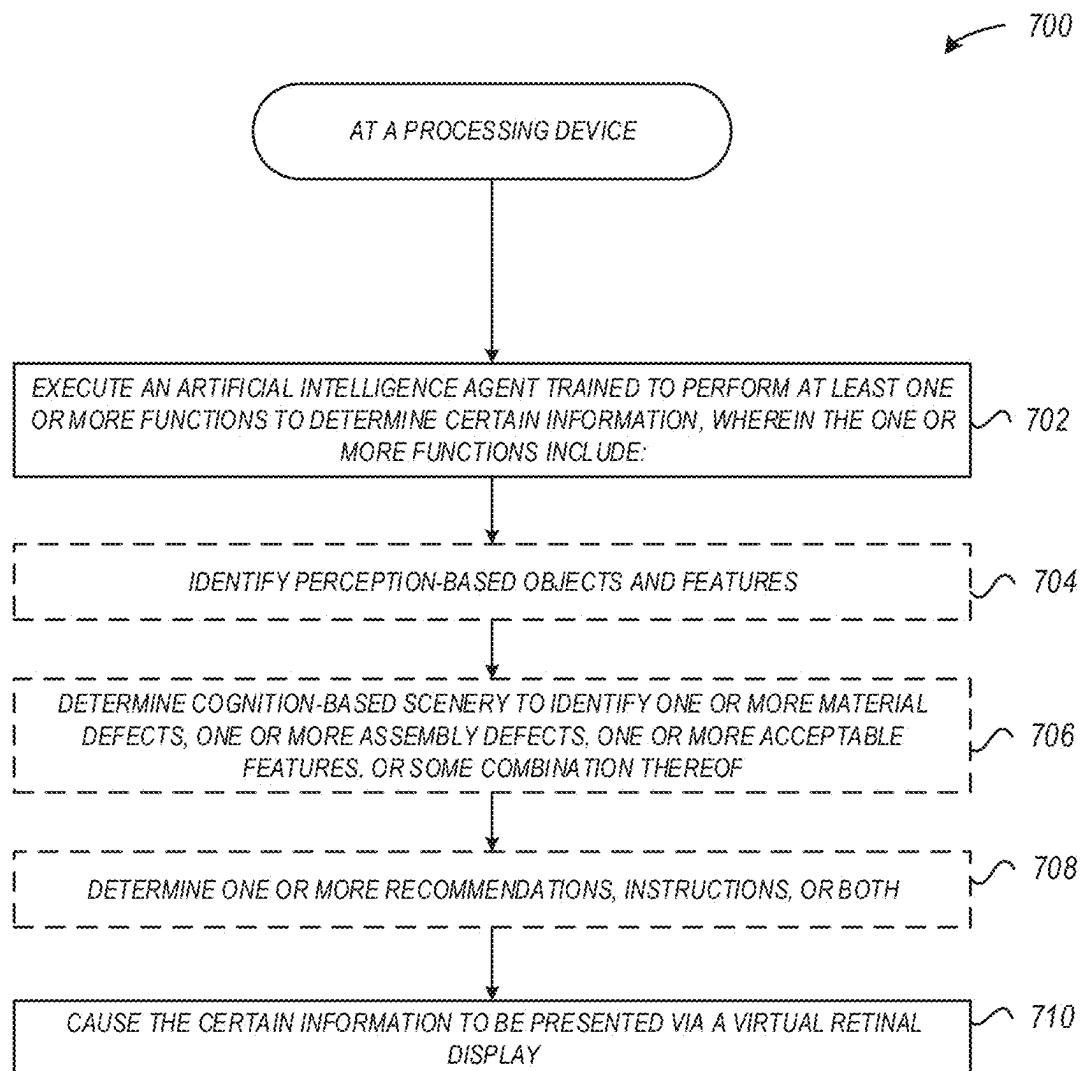


FIG. 7

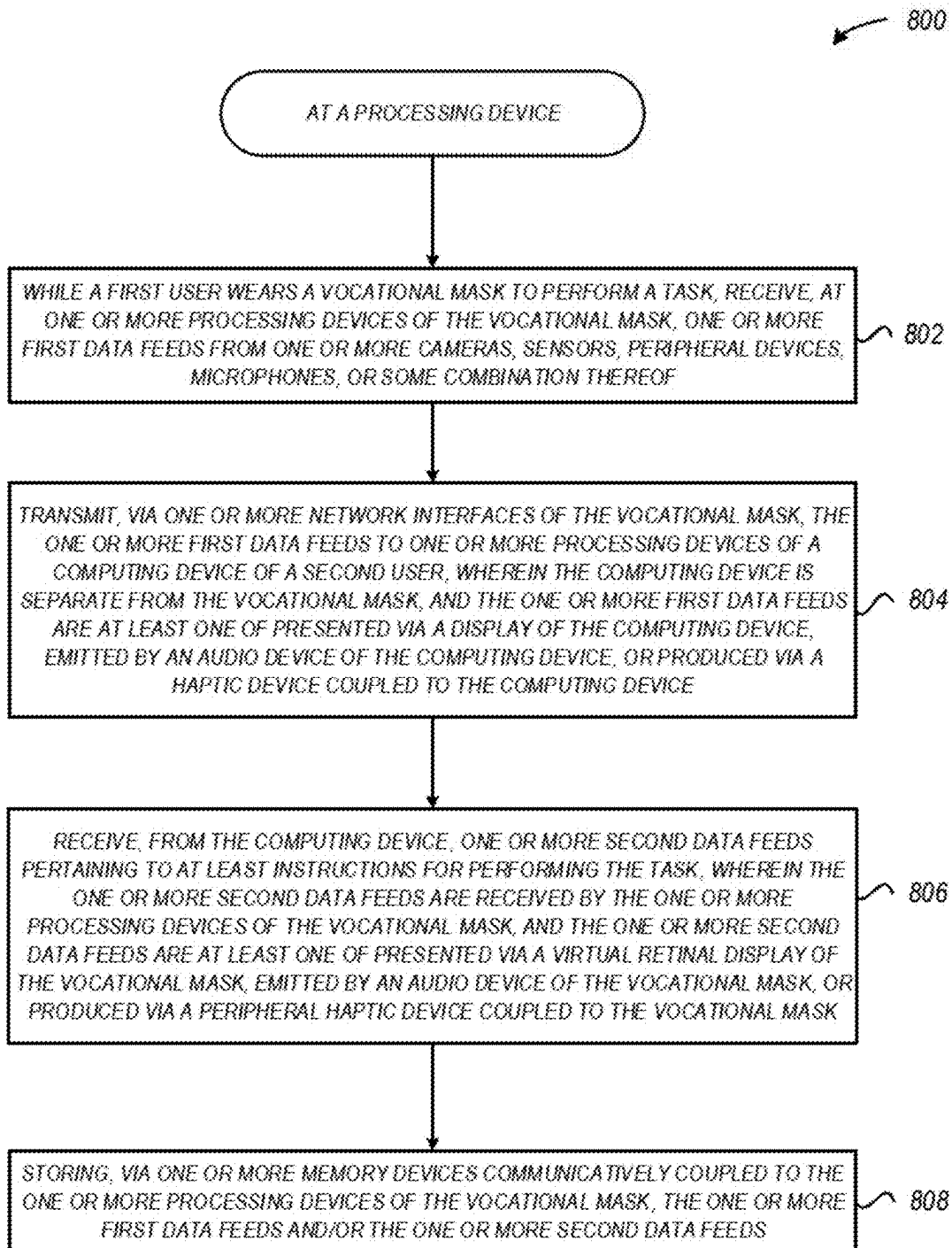


FIG. 8

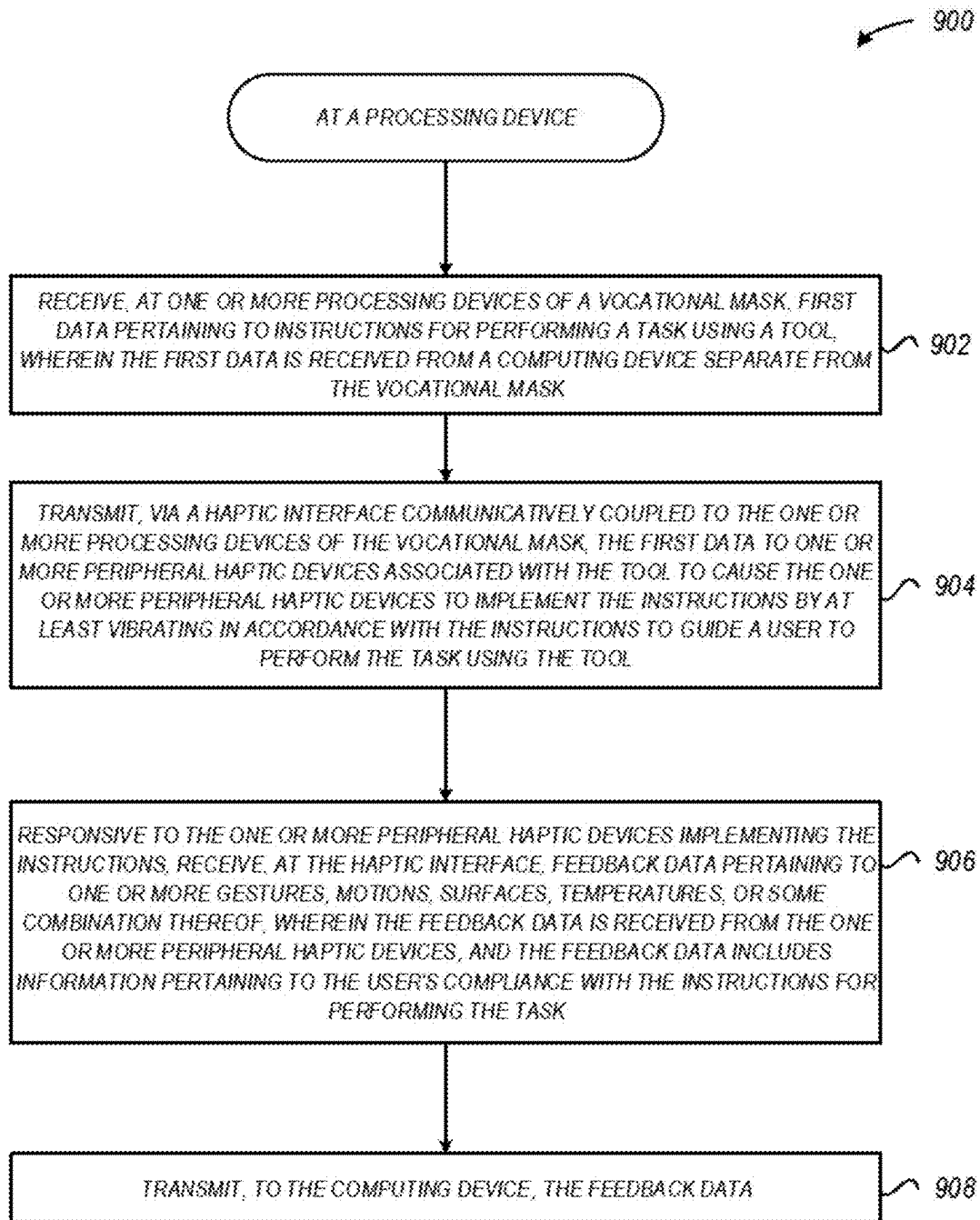


FIG. 9

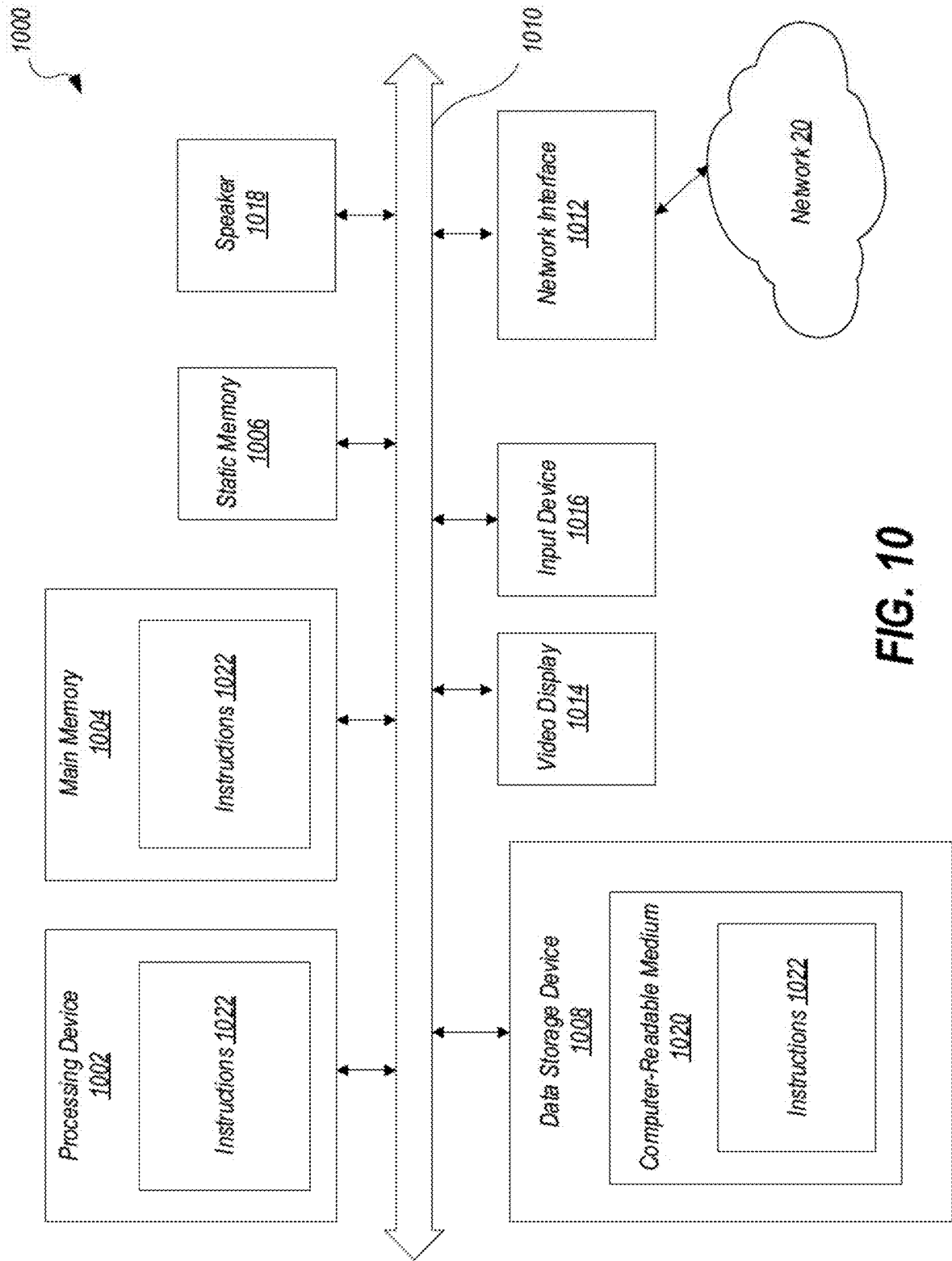


FIG. 10

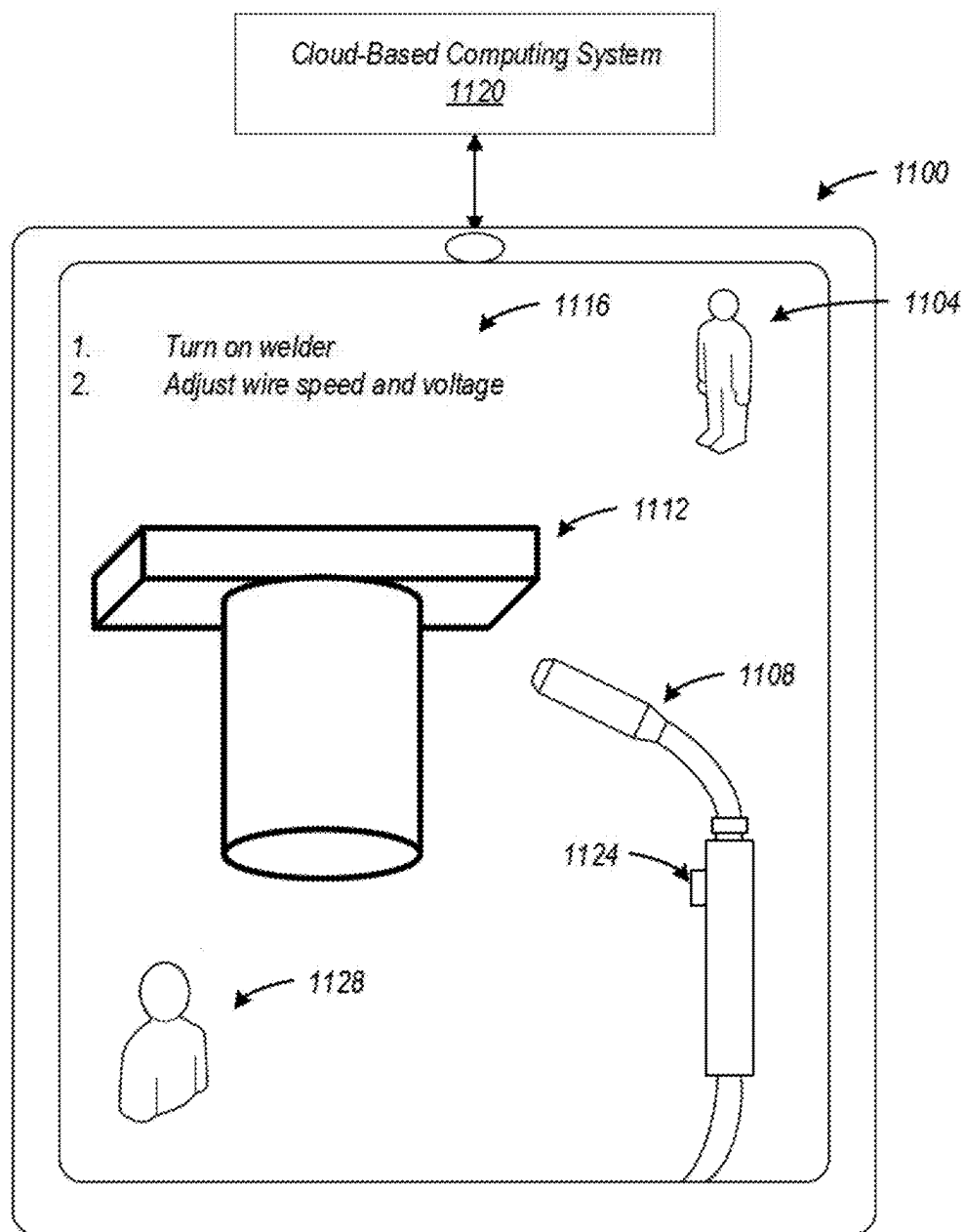


FIG. 11A

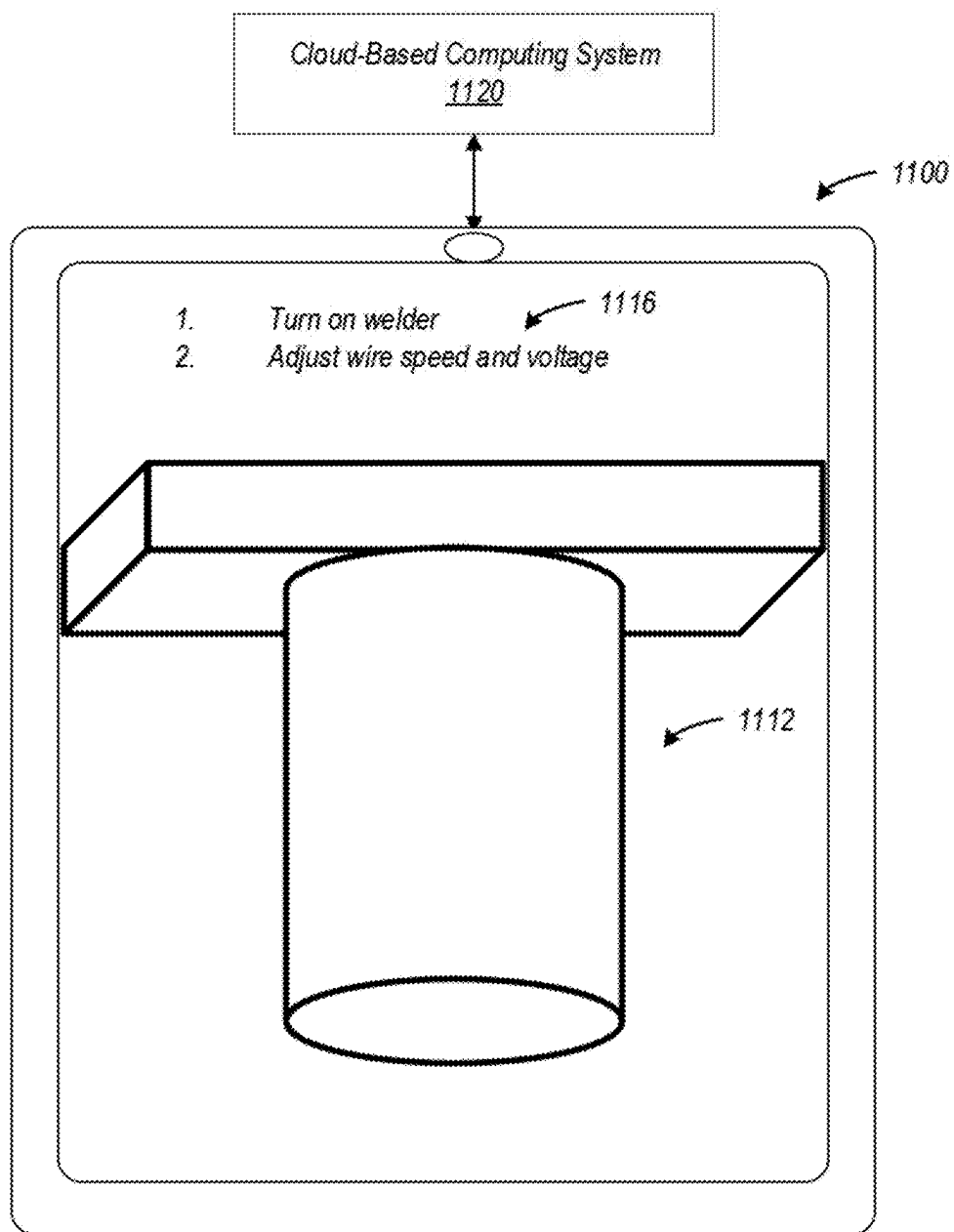


FIG. 11B

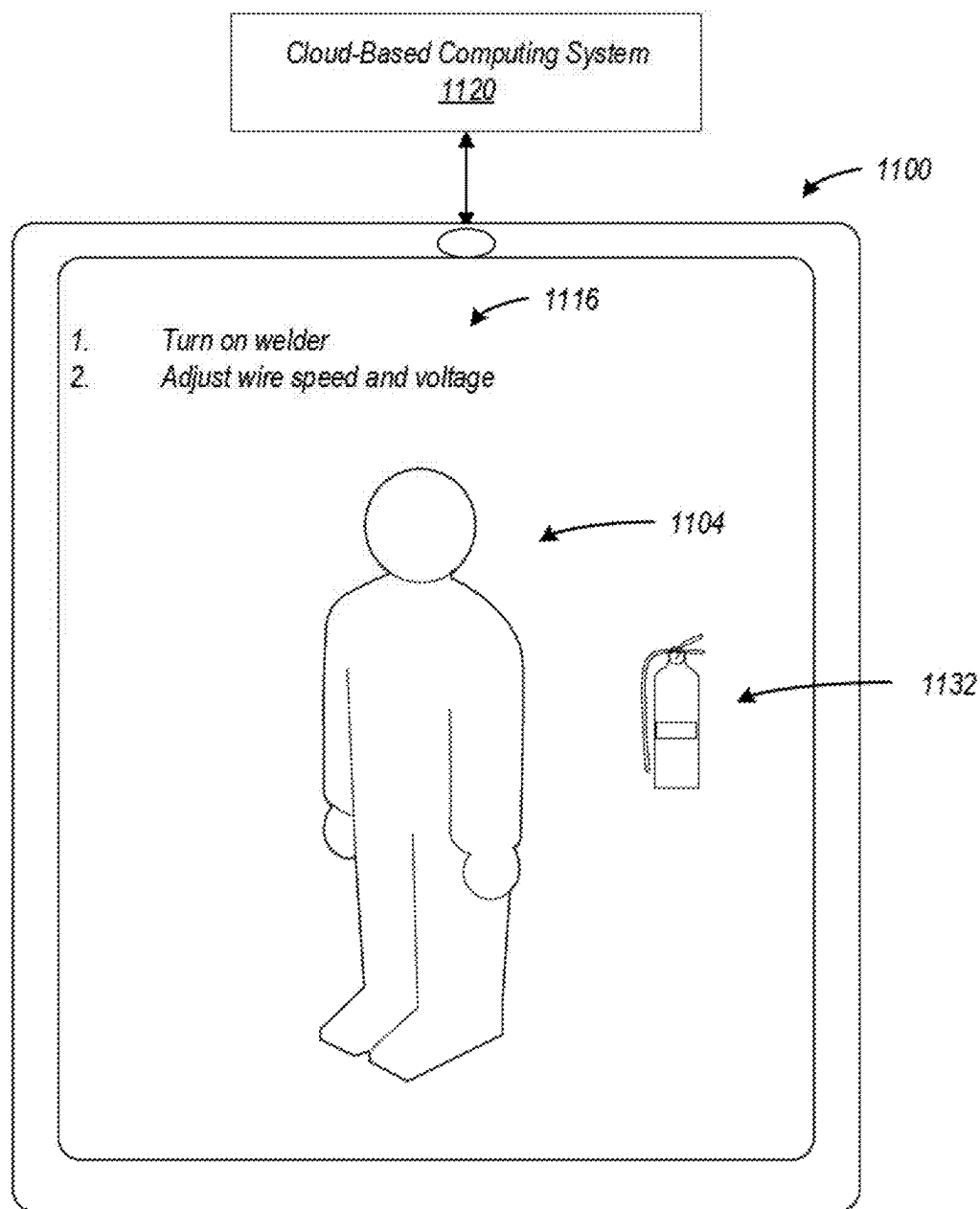


FIG. 11C

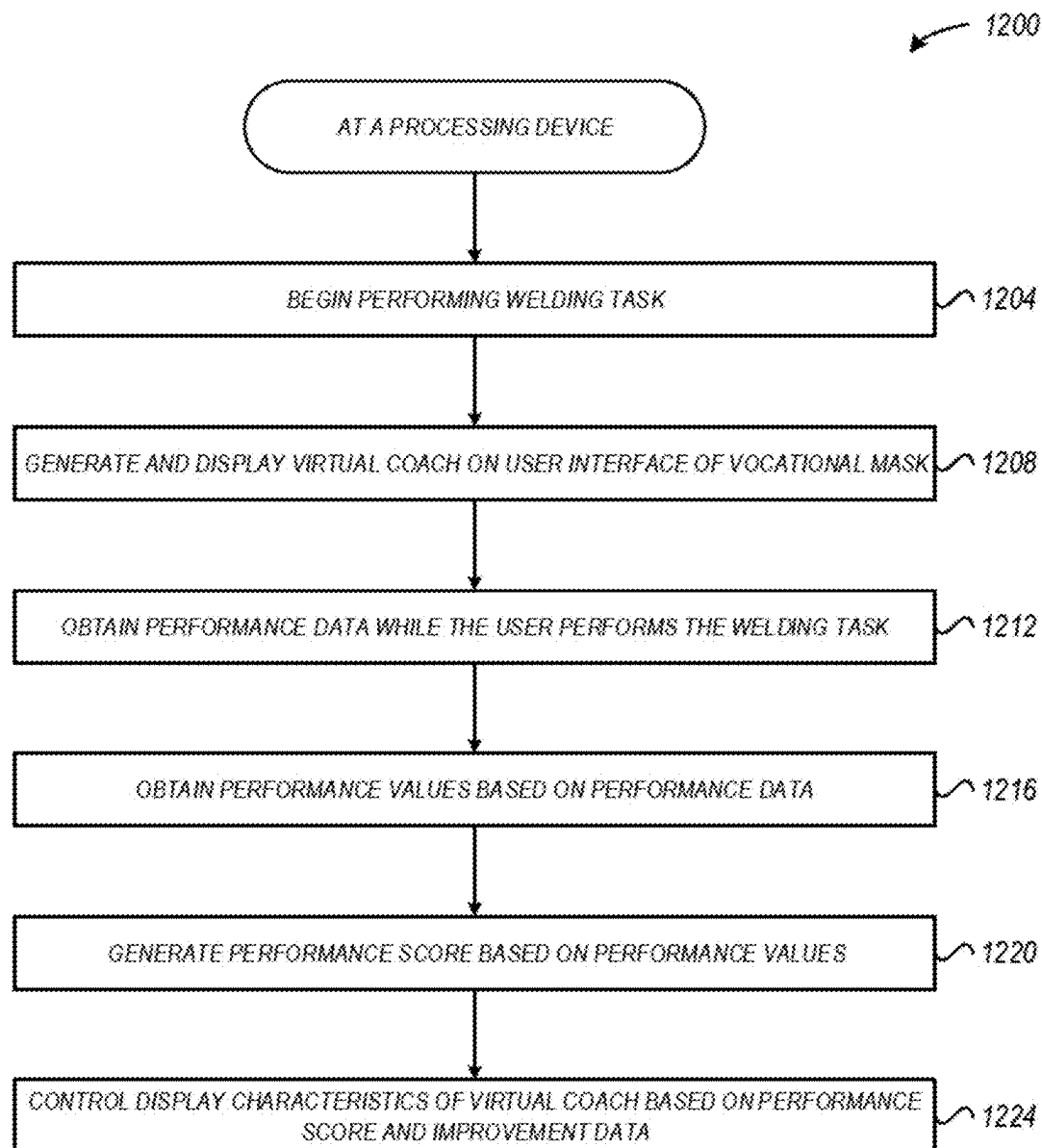


FIG. 12

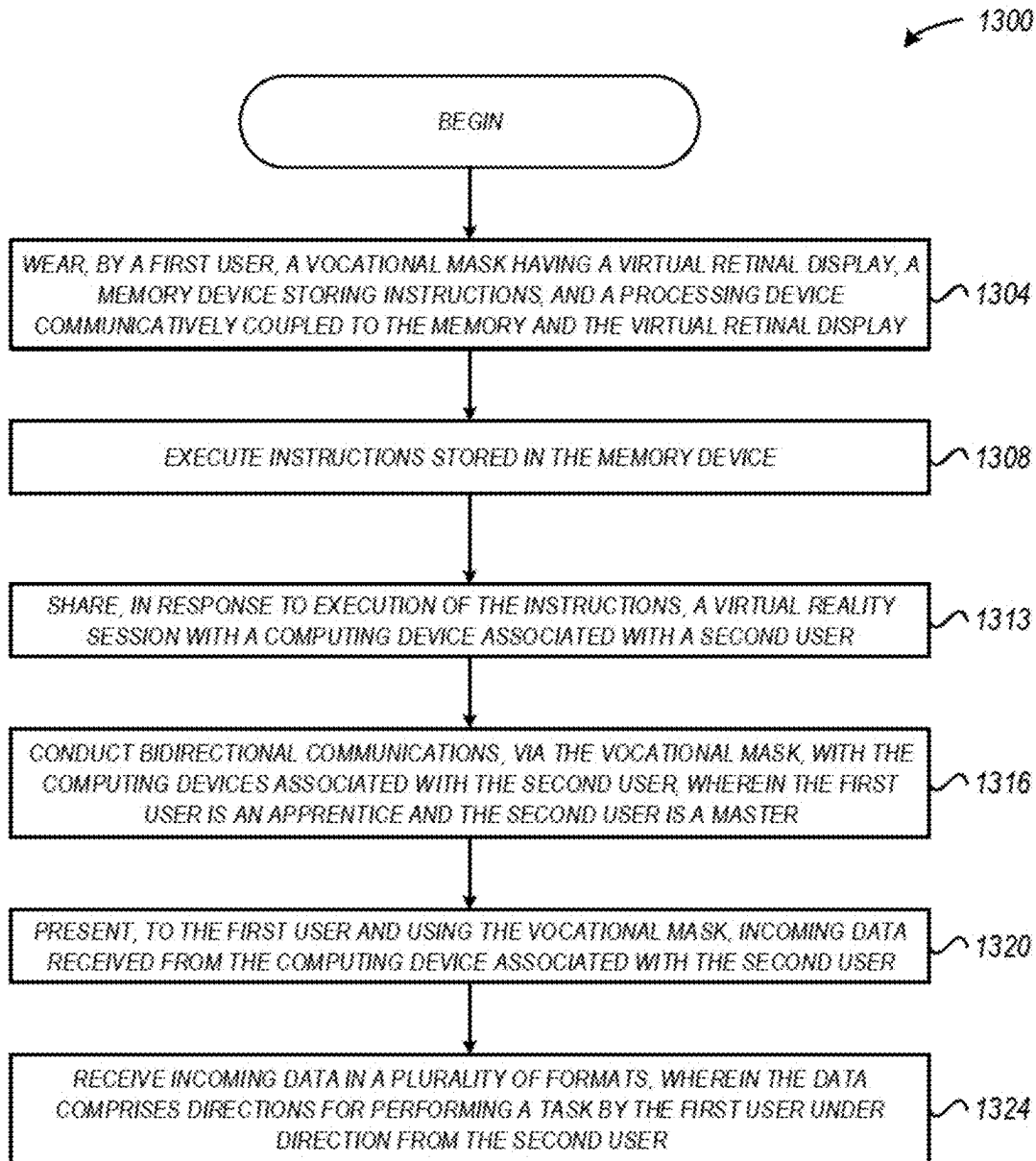


FIG. 13

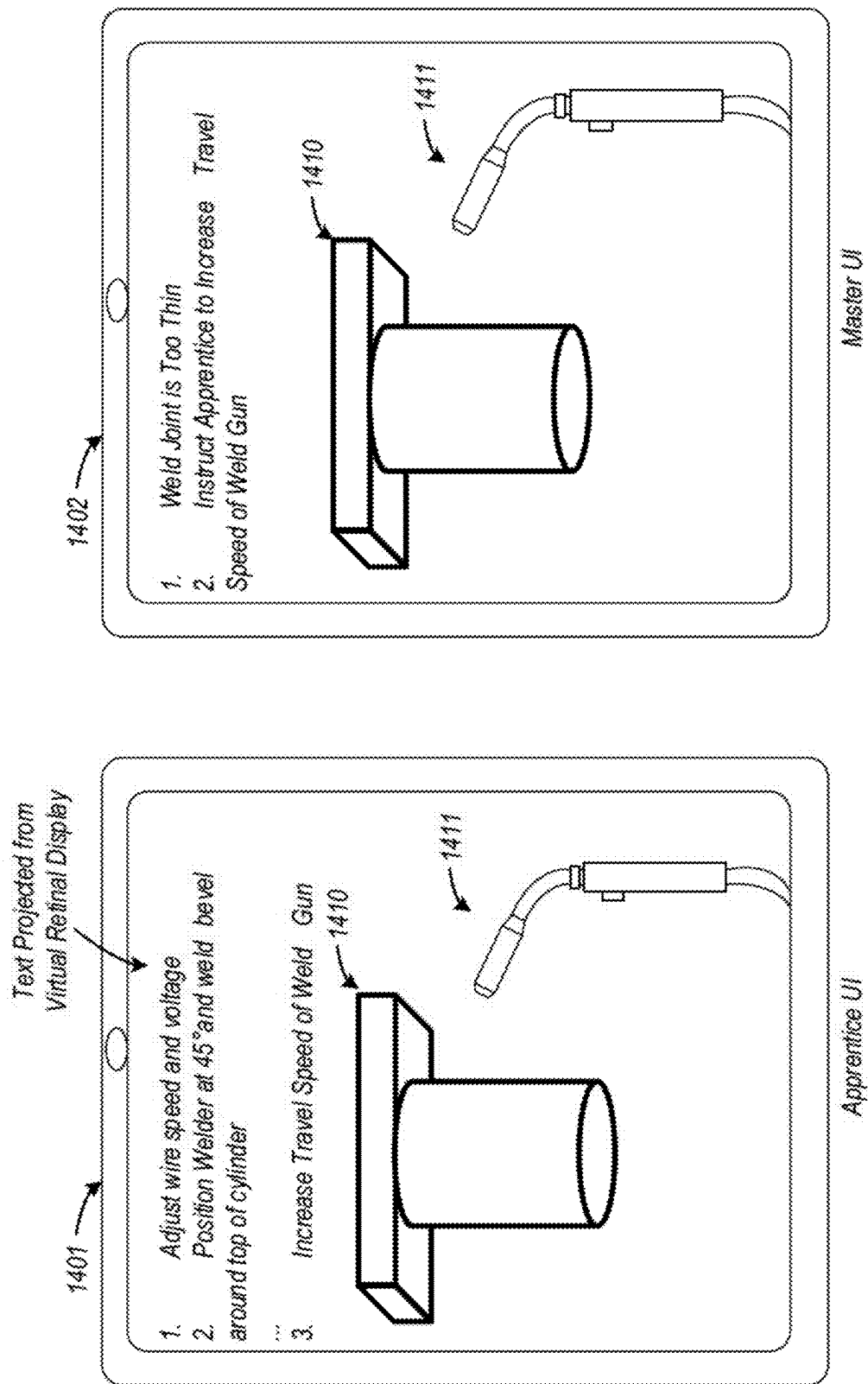


FIG. 14

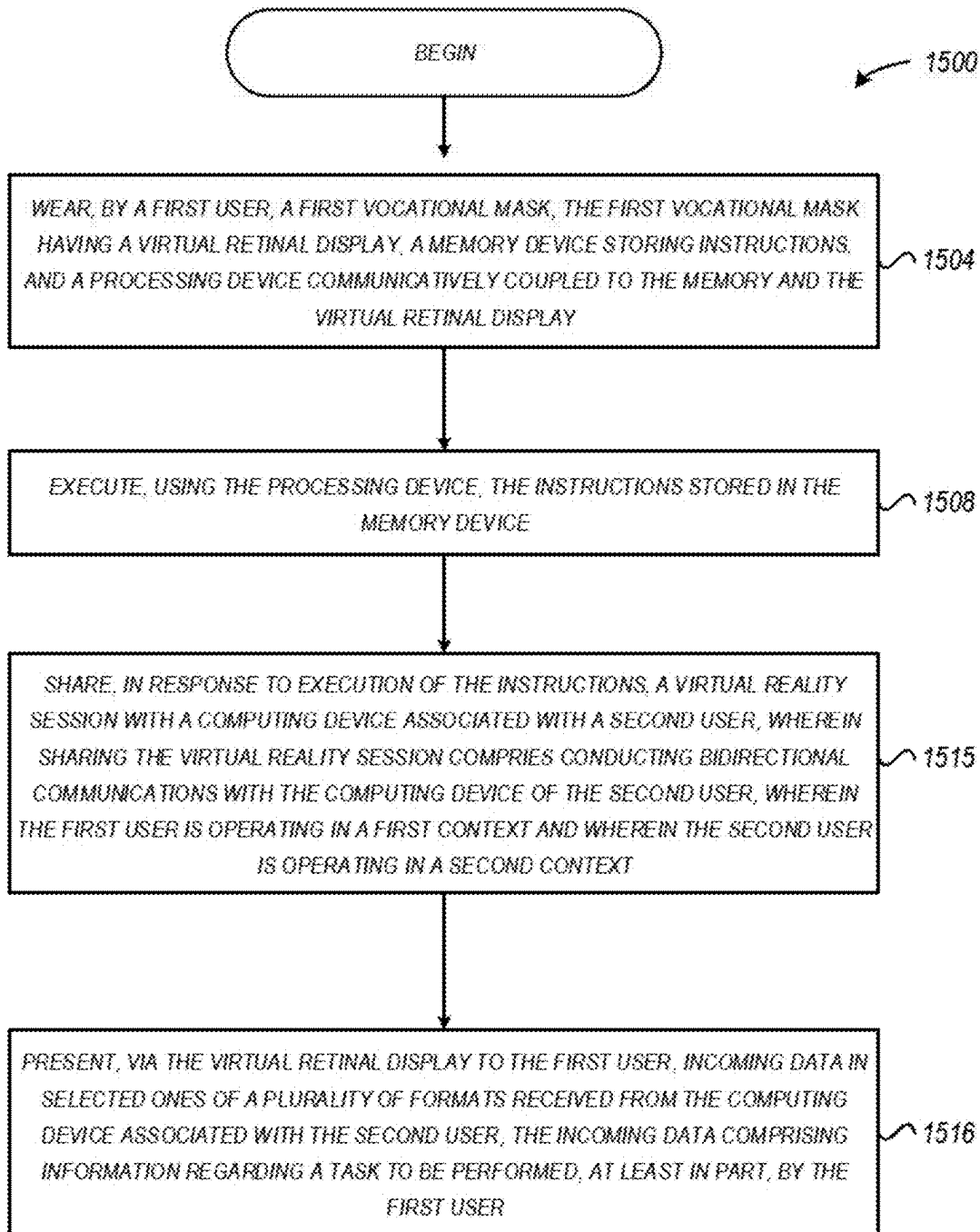


FIG. 15

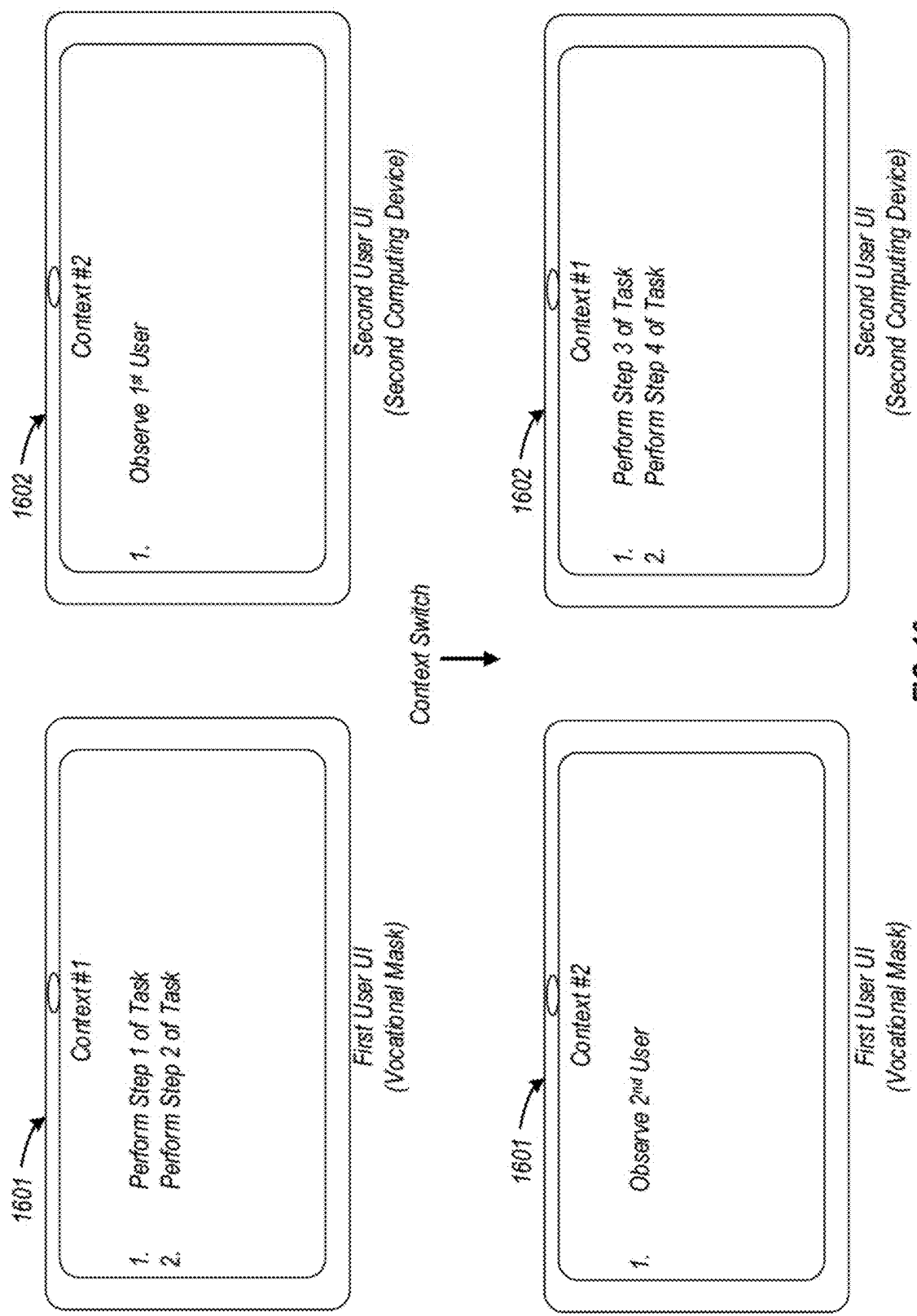


FIG. 16

**SYSTEMS AND METHODS FOR USING
VIRTUAL REALITY TO SIMULATE A WORK
TASK USING A VOCATIONAL MASK AND
BIDIRECTIONAL COMMUNICATION
BETWEEN AT LEAST TWO USERS**

**CROSS-REFERENCES TO RELATED
APPLICATIONS**

[0001] This application is a continuation-in-part of and claims the benefit of and priority to U.S. patent application Ser. No. 18/394,298 filed Dec. 22, 2023, titled “SYSTEMS AND METHODS FOR USING A VOCATIONAL MASK WITH A HYPER-ENABLED WORKER,” which claims priority to U.S. Patent Application Ser. No. 63/607,354 filed Dec. 7, 2023, titled “SYSTEMS AND METHODS FOR USING A VOCATIONAL MASK WITH A HYPER-ENABLED WORKER,” the entire disclosures of which are hereby incorporated by reference for all purposes.

[0002] This application also claims priority to U.S. Patent Application Ser. No. 63/633,166 filed Apr. 12, 2024, titled SYSTEMS AND METHODS FOR USING VIRTUAL REALITY TO SIMULATE A WORK TASK USING A VOCATIONAL MASK AND BIDIRECTIONAL COMMUNICATION BETWEEN AT LEAST TWO USERS,” the entire disclosure of which is hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0003] This disclosure relates to enabling workers to perform vocations. More specifically, this disclosure relates to systems and methods for using a vocational mask and a shared virtual reality session for instruction between a master and an apprentice.

BACKGROUND

[0004] People use various tools and/or equipment to perform various vocations. For example, a welder may use a welding mask and/or a welding gun to weld an object. The welder may participate in training courses prior to welding the object. A master welder may lead the training courses to train the welder how to properly weld. In some instances, the master welder may be located at a physical location that is remote from where a student welder is physically located.

SUMMARY

[0005] Systems and methods for using virtual reality to simulate a work task using a vocational mask and bidirectional communication between at least two users are disclosed. In one embodiment, a first vocational mask configured to be worn by a first user includes a virtual retinal display, a memory device storing instructions and a processing device communicatively coupled to the memory device and the virtual retinal display. The instructions, when executed by the processing device, are configured to cause the first vocational mask to share a virtual reality session with a computing device associated with a second user. During the virtual reality session, the first vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master. The first vocational mask is further configured to receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats from the second

user. The data comprises directions for performing a task by the first user under direction from the second user.

[0006] In one embodiment, a tangible, non-transitory computer-readable medium stores instructions that, when executed, cause a processing device to perform any operation of any method disclosed herein.

[0007] In one embodiment, a system includes a memory device storing instructions and a processing device communicatively coupled to the memory device. The processing device executes the instructions to perform any operation of any method disclosed herein.

[0008] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] For a detailed description of example embodiments, reference will now be made to the accompanying drawings in which:

[0010] FIG. 1 illustrates a system architecture according to certain embodiments of this disclosure;

[0011] FIG. 2 illustrates a component diagram for a vocational mask according to certain embodiments of this disclosure;

[0012] FIG. 3 illustrates bidirectional communication between communicatively coupled vocational masks according to certain embodiments of this disclosure;

[0013] FIG. 4 illustrates an example of projecting an image onto a user's retina via a virtual retinal display of a vocational mask according to certain embodiments of this disclosure;

[0014] FIG. 5 illustrates an example of an image including instructions projected via a virtual retinal display of a vocational mask according to certain embodiments of this disclosure;

[0015] FIG. 6 illustrates an example of an image including a warning projected via a virtual retinal display of a vocational mask according to certain embodiments of this disclosure;

[0016] FIG. 7 illustrates an example of a method for executing an artificial intelligence agent to determine certain information projected via a vocational mask of a user according to certain embodiments of this disclosure;

[0017] FIG. 8 illustrates an example of a method for transmitting instructions for performing a task via bidirectional communication between a vocational mask and a computing device according to certain embodiments of this disclosure;

[0018] FIG. 9 illustrates an example of a method for implementing instructions for performing a task using a peripheral haptic device according to certain embodiments of this disclosure;

[0019] FIG. 10 illustrates an example computer system according to embodiments of this disclosure;

[0020] FIGS. 11A-11C illustrate an example user interface configured to display a virtual avatar or coach according to embodiments of the present disclosure;

[0021] FIG. 12 illustrates steps of an example method of controlling the user interface and virtual coach of FIGS. 11A-11C according to embodiments of the present disclosure;

[0022] FIG. 13 illustrates steps of an example method of conducting a shared virtual reality session between an apprentice and a master according to embodiments of the present disclosure;

[0023] FIG. 14 illustrates examples of respective user interfaces for both an apprentice and a master during training to perform a particular task according to embodiments of the present disclosure;

[0024] FIG. 15 illustrates steps of an example of a method of conducting a shared virtual reality session between a first user and a second user operating in switchable first and second contexts according to embodiments of the present disclosure; and

[0025] FIG. 16 illustrates examples of respective user interfaces for first and second users sharing a virtual reality session with switchable contexts according to embodiments of the present disclosure.

NOTATION AND NOMENCLATURE

[0026] Various terms are used to refer to particular system components. Different entities may refer to a component by different names—this document does not intend to distinguish between components that differ in name but not function. In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to” Also, the term “couple” or “couples” is intended to mean either an indirect or a direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections.

[0027] The terminology used herein is for the purpose of describing particular example embodiments only, and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0028] The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections; however, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms, when used herein, do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C. In another example, the phrase

“one or more” when used with a list of items means there may be one item or any suitable number of items exceeding one.

[0029] Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), solid state drives (SSDs), flash memory, or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

[0030] Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

DETAILED DESCRIPTION

[0031] The following discussion is directed to various embodiments of the disclosed subject matter. Although one or more of these embodiments may be preferred, the embodiments disclosed should not be interpreted, or otherwise used, as limiting the scope of the disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to intimate that the scope of the disclosure, including the claims, is limited to that embodiment.

[0032] FIGS. 1 through 10 discussed below, and the various embodiments used to describe the principles of this disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure.

[0033] Some of the disclosed embodiments relate to one or more artificial intelligent enhanced vocational tools for workers to use to perform a job, task, and/or vocation. In some embodiments, the vocational tools may be in the form of a vocational mask that projects work instructions using imagery, animation, video, text, audio, and the like. The vocational tools may be used by workers to enhance the efficiency and proficiency of performing professional and vocational tasks, such as but not limited to supply chain operations, manufacturing and warehousing processes, product inspection, coworker and master-apprentice bidirectional collaboration and communication with or without haptic sensory feedback, other telepresence, and the like.

[0034] Some of the disclosed embodiments may be used to collect data, metadata, and multiband video to aid in product acceptance, qualification, and full lifecycle product management. Further, some of the disclosed embodiments may aid a failure reporting, analysis, and corrective action system, a failure mode, effects, and criticality analysis system, other sustainment and support activities and tasks to accommodate worker dislocation and multi-decade lifecycle of some products.

[0035] In one embodiment, a vocational mask is disclosed that employs bidirectional communication to include voice and imagery and still and audio video imagery recording with other colleagues over a distance. The vocational mask may provide virtual images of objects to a person wearing the vocational mask via a display (e.g., virtual retinal display). The vocational mask may enable bidirectional communications with collaborators and/or students (e.g., in a master-apprentice relationship). Further, the vocational mask may enable bidirectional audio, visual, and haptic communication to provide a master-apprentice relationship. A master-apprentice relationship may include virtually any teacher-student relationship or other type of relationship in which a wearer of the vocational mask carries out a task while receiving instructions or directions from another user via another computing device that is operatively coupled (e.g., via a network) to the vocational mask. Examples include, but are not limited to, an apprentice carrying out a task within a skilled trade under instruction of a master, a resident carrying out a medical procedure under instruction of a licensed doctor, and so on.

[0036] The disclosure further contemplates the wearer of the vocational mask being a collaborator with another user via another operatively coupled computing device (which may be another vocational mask or any other suitable computing device). Accordingly, two or more users may collaborate on performing a task or a learning exercise during a virtual reality session that is shared with or by a vocational mask.

[0037] Furthermore, the disclosure contemplates the sharing and switching of contexts between users in a shared virtual reality session, including at least one user wearing a vocational mask. For example a user wearing a first vocational mask may share a virtual reality session with another user wearing a second vocational mask, with the first user operating in a first context and the second user operating in a second context. However, these contexts may be switched during the virtual reality session such that, in this example, the first and second users exchange contexts. For example, in a master-apprentice relationship, a second user in the master role may send haptic data to a first user in an apprentice role to provide haptic instruction (e.g., vibration of a peripheral haptic device associated with a tool) on manipulating a tool. After a switch of contexts, the first user may send haptic data to the second user to enable the latter to use haptic feedback to monitor the manipulation of the tool by the first user.

[0038] In another example, two or more users performing a complex repair of a system may each, at times during the repair, require control of the system. A first user, wearing a vocational mask, may be on-site, and in a first context, may have control of the system under repair to enable certain steps to be carried out while the second user is in an observational context. At some point during the repair, a context switch may be performed in which the first user

transfers control of the system to be carried out remotely by the second user, with the first user assuming an observational role.

[0039] The vocational mask may include multiple electromagnetic spectrum and acoustic sensors/imagers. The vocational mask may also provide multiband audio and video sensed imagery to a wearer of the vocational mask.

[0040] The vocational mask may be configured to provide display capabilities to project images onto one or more irises of the wearer to display alphanumeric data and graphic/animated work instructions, for example. The vocational mask may also include one or more speakers to emit audio related to work instructions, such as those provided by a master trained user, supervisor, collaborator, teacher, etc.

[0041] The vocational mask may include an edge-based processor that executes an artificial intelligence agent. The artificial intelligence agent may be implemented in computer instructions stored on one or more memory devices and executed by one or more processing devices (e.g., edge processor or processors). The artificial intelligence agent may be trained to perform one or more functions, such as but not limited to (i) perception-based object and feature identification, (ii) cognition-based scenery understanding, to identify material and assembly defects versus acceptable features, and (iii) decision making to aid the wearer and to provide relevant advice and instruction in real-time or near real-time to the wearer of the vocational mask. The data that is collected may be used for inspection and future analyses of product quality, product design, and the like. Further, the collected data may be stored for instructional analyses and providing lessons, mentoring, collaboration, and the like.

[0042] The vocational mask may include one or more components (e.g., processing device, memory device, display, etc.), interfaces, and/or sensors configured to provide sensing capabilities to understand hand motions and use of a virtual user interface (e.g., keyboards) and other haptic instructions. The vocational mask may include a haptic interface to allow physical bidirectional haptic sensing and stimulation via the bidirectional communications to other users and/or collaborators using a peripheral haptic device (e.g., a welding gun).

[0043] In some embodiments, the vocational mask may be in the form of binocular goggles, monocular goggle, finishing process glasses (e.g., grind, chamfer, debur, sand polish, coat, etc.), or the like. The vocational mask may be attached to a welding helmet. The vocational mask may include an optical bench that aligns a virtual retinal display to one or more eyes of a user. Certain types of data (e.g., video and image) may be projected from the virtual retinal display into the retinal of a user. The vocational mask may include a liquid crystal display welding helmet, a welding camera, an augmented reality/virtual reality headset, etc.

[0044] The vocational mask may augment projections by providing augmented reality cues and information to assist a worker (e.g., welder) with contextual information, which may include setup, quality control, procedures, training, and the like. Further, the vocational mask may provide a continuum of visibility from visible spectrum (arc off) through high-intensity/ultraviolet (arc on). Further, some embodiments include remote feedback and recording of images and bidirectional communications to a trainer, supervisor, mentor, master user, teacher, collaborator, etc. who can provide visual, auditory, and/or haptic feedback to the wearer of the vocational mask in real-time or near real-time.

[0045] In some embodiments, the vocational mask may be integrated with a welding helmet. In some embodiments, the vocational mask may be (or include) a set of augmented reality/virtual reality goggles worn under a welding helmet (e.g., with external devices, sensors, cameras, etc. appended for image/data gathering). In some embodiments, the vocational mask may be a set of binocular welding goggles or a monocular welding goggle to be worn under or in lieu of a welding helmet (e.g., with external devices, sensors, cameras, etc. appended to the goggles for image/data gathering). In some embodiments, the vocational mask may include a mid-band or long wave context camera displayed to the user and monitor.

[0046] In some embodiments, information may be superpositioned or superimposed onto a display without the user (e.g., worker, student, etc.) wearing a vocational mask. The information may include work instructions in the form of text, images, alphanumeric characters, video, etc. The vocational mask may function across both visible light (arc off) and high intensity ultraviolet light (arc on) conditions. The vocational mask may natively or in conjunction with other personal protective equipment provide protection against welding flash. The vocational mask may enable real-time or near real-time two-way communication with a remote instructor or supervisor. The vocational mask may provide one or more video, audio, and data feeds to a remote instructor or supervisor. The vocational mask and/or other components in a system may enable recording of all data and communications. The system may provide a mechanism for replaying the data and communications, via a media player, for training purposes, quality control purposes, inspection purposes, and the like. The vocational mask and/or other components in a system may provide a mechanism for visual feedback from a remote instructor or supervisor. The vocational mask and/or other components in a system may provide a bidirectional mechanism for haptic feedback from a remote instructor or supervisor.

[0047] Further, the system may include an artificial intelligence simulation generator that generates task simulations to be transmitted to and presented via the vocational mask. The simulation of a task may be transmitted as virtual reality data to the vocational mask which includes a virtual reality headset and/or display to playback the virtual reality data. The virtual reality data may be configured based on parameters of a physical space in which the vocational mask is located, based on parameters of an object to be worked on, based on parameters of a tool to be used, and the like.

[0048] Some embodiments of the system may also include an artificial intelligence agent that is implemented in instructions stored on one or more memory devices and executable on one or more processing devices of the vocational mask. The artificial intelligence agent is trained such that, when executed, it may monitor one or more aspects of the virtual reality session and may additionally provide directions for performing a task to a user wearing the vocational mask. The artificial intelligence agent may also monitor one or more properties of a task performed by a user wearing the vocational mask. For example, if the task is welding, the artificial intelligence agent may monitor one or more properties of a weld formed by the first user. Based on the one or more monitored properties, the artificial intelligence agent may adjust the directions provided to the user for carrying out the task. In various embodiments, the artificial intelligence agent may monitor a number of different types of tasks in

addition to the example of welding given here. Other tasks monitored by the artificial intelligence agent may include (but are not limited to) brazing, soldering, and other types of mechanical and/or industrial processes that may be carried out by a user, medical procedures to be carried out by a resident under instruction of a doctor (as well as procedures carried out by one doctor with the assistance of another), repair operations carried out by a technician with the assistance of an engineer or other technician, and so on.

[0049] Turning now to the figures, FIG. 1 depicts a system architecture 10 according to some embodiments. The system architecture 10 may include one or more computing devices 140, one or more vocational masks 130, one or more peripheral haptic devices 134, and/or one or more tools 136 communicatively coupled to a cloud-based computing system 116. Each of the computing devices 140, vocational masks 130, peripheral haptic devices 134, tools 136, and components included in the cloud-based computing system 116 may include one or more processing devices, memory devices, and/or network interface cards. The network interface cards may enable communication via a wireless protocol for transmitting data over short distances, such as Bluetooth, ZigBee, NFC, etc. Additionally, the network interface cards may enable communicating data over long distances, and in one example, the computing devices 140, the vocational masks 130, the peripheral haptic devices 134, the tools 136, and the cloud-based computing system 116 may communicate with a network 20. Network 20 may be a public network (e.g., connected to the Internet via wired (Ethernet) or wireless (WiFi)), a private network (e.g., a local area network (LAN) or wide area network (WAN)), or a combination thereof. Network 20 may also include a node or nodes on the Internet of Things (IoT). The network 20 may be a cellular network.

[0050] The computing devices 140 may be any suitable computing device, such as a laptop, tablet, smartphone, smartwatch, ear buds, server, or computer. In some embodiments, the computing device 140 may be a vocational mask. The computing devices 140 may include a display capable of presenting a user interface 142 of an application. In some embodiments, the display may be a laptop display, smartphone display, computer display, tablet display, a virtual retinal display, etc. The application may be implemented in computer instructions stored on the one or more memory devices of the computing devices 140 and executable by the one or more processing devices of the computing device 140. The application may present various screens to a user. For example, the user interface 142 may present a screen that plays video received from the vocational mask 130. The video may present real-time or near real-time footage of what the vocational mask 130 is viewing, and in some instances, that may include a user's hands holding the tool 136 to perform a task (e.g., weld, sand, polish, chamfer, debur, paint, play a video game, etc.). Additional screens may be presented via the user interface 142.

[0051] In some embodiments, the application (e.g., website) executes within another application (e.g., web browser). The computing device 140 may also include instructions stored on the one or more memory devices that, when executed by the one or more processing devices of the computing devices 140 perform operations of any of the methods described herein.

[0052] In some embodiments, the computing devices 140 may include an edge processor 132.1 that performs one or

more operations of any of the methods described herein. The edge processor **132.1** may execute an artificial intelligence agent to perform various operations described herein. The artificial intelligence agent may include one or more machine learning models that are trained via the cloud-based computing system **116** as described herein. The edge processor **132.1** may represent one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. More particularly, the edge processor **132.1** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets or processors implementing a combination of instruction sets. The edge processor **132.1** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like.

[0053] In some embodiments, the vocational mask **130** may be attached to or integrated with a welding helmet, binocular goggles, a monocular goggle, glasses, a hat, a helmet, a virtual reality headset, a headset, a facemask, or the like. The vocational mask **130** may include various components as described herein, such as an edge processor **132.2**. In some embodiments, the edge processor **132.2** may be located separately from the vocational mask **130** and may be included in another computing device, such as a server, laptop, desktop, tablet, smartphone, etc. The edge processor **132.2** may represent one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. More particularly, the edge processor **132.2** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets or processors implementing a combination of instruction sets. The edge processor **132.2** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like.

[0054] The edge processor **132.2** may perform one or more operations of any of the methods described herein. The edge processor **132.2** may execute an artificial intelligence agent to perform various operations described herein. The artificial intelligence agent may include one or more machine learning models that are trained via the cloud-based computing system **116** as described herein. For example, the cloud-based computing system **116** may train one or more machine learning models **154** via a training engine **152**, and may transmit the parameters used to train the machine learning model to the edge processor **132.2** such that the edge processor **132.2** can implement the parameters in the machine learning models executing locally on the vocational mask **130** or computing device **140**. The machine learning models may be trained to monitor various aspects of a virtual reality session between, e.g., an apprentice and a master, and may further be trained to monitor properties of tasks carried out by the apprentice (e.g., a weld, a solder joint, etc.). Based on this monitoring, an artificial intelligence agent using these machine learning models may adjust instructions/directions provided to the apprentice.

[0055] The edge processor **132.2** may include a data concentrator that collects data from multiple vocational masks **130** and transmits the data to the cloud-based computing system **116**. The data concentrator may map information to reduce bandwidth transmission costs of transmitting data. In some embodiments, a network connection may not be needed for the edge processor **132.2** to collect data from vocational masks and to perform various functions using the trained machine learning models **154**.

[0056] The vocational mask **130** may also include a network interface card that enables bidirectional communication with any other computing device **140**, such as other vocational masks **130**, smartphones, laptops, desktops, servers, wearable devices, tablets, etc. The vocational mask **130** may also be communicatively coupled to the cloud-based computing system **116** and may transmit and receive information and/or data to and from the cloud-based computing system **116**. The vocational mask **130** may include various sensors, such as position sensors, acoustic sensors, haptic sensors, microphones, temperature sensors, accelerometers, and the like. The vocational mask **130** may include various cameras configured to capture audio and video. The vocational mask **130** may include a speaker to emit audio. The vocational mask **130** may include a haptic interface configured to transmit and receive haptic data to and from the peripheral haptic device **134**. The haptic interface may be communicatively coupled to a processing device (e.g., edge processor **132.2**) of the vocational mask **130**.

[0057] In some embodiments, the peripheral haptic device **134** may be attached to or integrated with the tool **136**. In some embodiments, the peripheral haptic device **134** may be separate from the tool **136**. The peripheral haptic device **134** may include one or more haptic sensors that provide force, vibration, touch, and/or motion sensations to the user, among other things. The peripheral haptic device **134** may be used to enable a person remote from a user of the peripheral haptic device **134** to provide haptic instructions to perform a task (e.g., weld, shine, polish, paint, control a video game controller, grind, chamfer, debur, etc.). The peripheral haptic device **134** may include one or more processing devices, memory devices, network interface cards, haptic interfaces, etc. In some embodiments, the peripheral haptic device **134** may be communicatively coupled to the vocational mask **130**, the computing device **140**, and/or the cloud-based computing system **116**.

[0058] The tool **136** may be any suitable tool, such as a welding gun, a video game controller, a paint brush, a pen, a utensil, a grinder, a sander, a polisher, a gardening tool, a yard tool, a glove, medical tools, a brazing torch, a soldering iron, a mechanical tool such as a wrench or pliers, or the like. The tool **136** may be handheld such that the peripheral haptic device **134** is enabled to provide haptic instructions for performing a task to the user holding the tool **136**. In some embodiments, the tool **136** may be wearable by the user. The tool **136** may be used to perform a task. In some embodiments, the tool **136** may be located in a physical proximity to the user in a physical space.

[0059] In some embodiments, the cloud-based computing system **116** may include one or more servers **128** that form a distributed computing architecture. The servers **128** may be a rackmount server, a router computer, a personal computer, a portable digital assistant, a mobile phone, a laptop computer, a tablet computer, a camera, a video camera, a netbook, a desktop computer, a media center, any other

device capable of functioning as a server, or any combination of the above. Each of the servers **128** may include one or more processing devices, memory devices, data storage, and/or network interface cards. The servers **128** may be in communication with one another via any suitable communication protocol. The servers **128** may execute an artificial intelligence (AI) engine and/or an AI agent that uses one or more machine learning models **154** to perform at least one of the embodiments disclosed herein. The cloud-based computing system **116** may also include a database **129** that stores data, knowledge, and data structures used to perform various embodiments. For example, the database **129** may store multimedia data of users performing tasks using tools, communications between vocational masks **130** and/or computing devices **140**, virtual reality simulations, augmented reality information, recommendations, instructions, and the like. The database **129** may also store user profiles including characteristics particular to each user. In some embodiments, the database **129** may be hosted on one or more of the servers **128**.

[0060] In some embodiments the cloud-based computing system **116** may include a training engine **152** capable of generating the one or more machine learning models **154**. The machine learning models **154** may be trained to identify perception-based objects and features using training data that includes labeled inputs of images including certain objects and features mapped to labeled outputs of identities or characterizations of those objects and features. The machine learning models **154** may be trained to determine cognition-based scenery to identify one or more material defects, one or more assembly defects, one or more acceptable features, or some combination thereof using training data that includes labeled input of scenery images of objects including material defects, assembly defects, and/or acceptable features mapped to labeled outputs that characterize and/or identify the material defects, assembly defects, and/or acceptable features. The machine learning models **154** may be trained to determine one or more recommendations, instructions, or both using training data including labeled input of images (e.g., objects, products, tools, actions, etc.) and tasks to be performed (e.g., weld, grind, chamfer, debur, sand, polish, coat, etc.) mapped to labeled outputs including recommendations, instructions, or both.

[0061] The one or more machine learning models **154** may be generated by the training engine **152** and may be implemented in computer instructions executable by one or more processing devices of the training engine **152** and/or the servers **128**. To generate the one or more machine learning models **154**, the training engine **152** may train the one or more machine learning models **154**. The one or more machine learning models **154** may also be executed by the edge processor **132** (**132.1**, **132.2**). The parameters used to train the one or more machine learning models **154** by the training engine **152** at the cloud-based computing system **116** may be transmitted to the edge processor **132** (**132.1**, **132.2**) to be implemented locally at the vocational mask **130** and/or the computing device **140**.

[0062] The training engine **152** may be a rackmount server, a router computer, a personal computer, a portable digital assistant, a smartphone, a laptop computer, a tablet computer, a netbook, a desktop computer, an Internet of Things (IoT) device, any other desired computing device, or any combination of the above. The training engine **152** may be cloud-based, be a real-time software platform, include

privacy software or protocols, and/or include security software or protocols. To generate the one or more machine learning models **154**, the training engine **152** may train the one or more machine learning models **154**.

[0063] The one or more machine learning models **154** may refer to model artifacts created by the training engine **152** using training data that includes training inputs and corresponding target outputs. The training engine **152** may find patterns in the training data wherein such patterns map the training input to the target output and generate the machine learning models **154** that capture these patterns. Although depicted separately from the server **128**, in some embodiments, the training engine **152** may reside on server **128**. Further, in some embodiments, the database **129**, and/or the training engine **152** may reside on the computing devices **140**.

[0064] As described in more detail below, the one or more machine learning models **154** may comprise, e.g., a single level of linear or non-linear operations (e.g., a support vector machine [SVM]) or the machine learning models **154** may be a deep network, i.e., a machine learning model comprising multiple levels of non-linear operations. Examples of deep networks are neural networks, including generative adversarial networks, convolutional neural networks, recurrent neural networks with one or more hidden layers, and fully connected neural networks (e.g., each neuron may transmit its output signal to the input of the remaining neurons, as well as to itself). For example, the machine learning model may include numerous layers and/or hidden layers that perform calculations (e.g., dot products) using various neurons.

[0065] FIG. 2 illustrates a component diagram for a vocational mask **130** according to certain embodiments of this disclosure. The edge processor **132.2** is also depicted. In some embodiments, the edge processor **132.2** may be included in a computing device separate from the vocational mask **130**, and in some embodiments, the edge processor **132.2** may be included in the vocational mask **130**.

[0066] The vocational mask **130** may include various position, navigation, and time (PNT) components, sensors, and/or devices that enable determining the geographical position (latitude, longitude, altitude, time), pose (length (ground to sensor), elevation, time), translation (delta in latitude, delta in longitude, delta in altitude, time), the rotational rate of pose ((ω_r , ω_p , ω_y (northing), t)), and the like, where ω_r represents the roll rate, which is the angular velocity about the longitudinal axis of the vocational mask **130**, ω_p represents the pitch rate, which is the angular velocity about the lateral axis of the vocational mask **130**, ω_y (northing) represents the yaw rate, which is the angular velocity about the vertical axis of the vocational mask **130**, references with respect to the northing direction, and t represents the time at which these rotational rates are measured.

[0067] In some embodiments, the vocational mask **130** may include one or more sensors, such as vocation imaging band specific cameras, visual band cameras, microphones, and the like.

[0068] In some embodiments, the vocational mask **130** may include an audio visual display, such as a stereo speaker, a virtual retinal display, a liquid crystal display, a virtual reality headset, and the like. A virtual retinal display may be a retinal scan display or retinal projector that draws a raster display directly onto the retina of the eye. In some

embodiments, the virtual retinal display may include drive electronics that transmit data to a photon generator and/or intensity modulator. These components may process the data (e.g., video, audio, haptic, etc.) and transmit the processed data to a beam scanning component that further transmits data to an optical projector that projects an image and/or video to a retina of a user.

[0069] In some embodiments, the vocational mask 130 may include a network interface card that enables bidirectional communication (digital communication) with other vocational masks and/or computing device 140.

[0070] In some embodiments, the vocational mask 130 may provide a user interface to the user via the display described herein.

[0071] In some embodiments, the edge processor 132.2 may include a network interface card that enables digital communication with the vocational mask 130, the computing device 140, the cloud-based computing system 116, or the like.

[0072] FIG. 3 illustrates bidirectional communication between communicatively coupled vocational masks 130 and 302 according to certain embodiments of this disclosure. As depicted, a user 306 is wearing a vocational mask 130. In the depicted example, the vocational mask 130 is attached to or integrated with a welding helmet 308. The user is viewing an object or workpiece 300. The vocational mask 130 may include multiple electromagnetic spectrum and/or acoustic sensors/imagers 304 to enable obtaining audio, video, acoustic, etc. data while observing the object 300 and/or performing a task (e.g., welding).

[0073] Further, as depicted, the vocational mask 130 may be communicatively coupled to one or more other vocational masks 302 worn by other users and may communicate data in real-time or near real-time such that bidirectional audio visual and haptic communications fosters a master-apprentice relationship, a teacher-student relationship, a relationship between collaborators, or other type of relationship. In some embodiments, the bidirectional communication enabled by the vocational masks 130 may enable collaboration between a teacher or collaborator and students. Each of the users wearing the vocational mask 130 may be enabled to visualize the object 300 that the user is viewing in real-time or near real-time.

[0074] FIG. 4 illustrates an example of projecting an image onto a user's retina 400 via a virtual retinal display of a vocational mask 130 according to certain embodiments of this disclosure. As depicted, the imagers and/or cameras of the vocational mask 130 receive data pertaining to the object and the vocational mask 130 processes the data and projects an image representing the object 300 using a virtual retinal display onto the user's retina 400. The bidirectional communication with other users (e.g., students, master user, collaborator, teacher, supervisor, etc.) may enable projecting the image onto their retinas if they are wearing a vocational mask, as well. In some embodiments, the image may be displayed via a computing device 140 if the other users are not wearing vocational masks.

[0075] FIG. 5 illustrates an example of an image including instructions projected via a virtual retinal display of a vocational mask 130 according to certain embodiments of this disclosure. The example user interface 500 depicts actual things the user is looking at, such as a tool 136 and an object 300 (e.g., a workpiece), through the vocational mask 130. Further, the user interface depicts instructions 502

pertaining to performing a task. The instructions 502 may be generated by one or more machine learning models 154 of the AI agent, or may be provided via a computing device 140 and/or other vocational mask being used by another user (e.g., master user, collaborator, teacher, supervisor, etc.). In the depicted example, the instructions 502 instruct the user to "1. Turn on welder; 2. Adjust wire speed and voltage". The instructions 502 may be projected on the user's retina via the virtual retinal display and/or presented on a display of the vocational mask 130.

[0076] FIG. 6 illustrates an example of an image including a warning projected via a virtual retinal display of a vocational mask according to certain embodiments of this disclosure. The example user interface 600 depicts actual things the user is looking at, such as a tool 136 and an object 300, through the vocational mask 130. Further, the user interface depicts a warning 602 pertaining to performing a task. The warning 602 may be generated by one or more machine learning models 154 of the AI agent, or may be provided via a computing device 140 and/or other vocational mask being used by another user (e.g., master user, collaborator, teacher, supervisor, etc.). In the depicted example, the warning 602 indicates "Caution: Material defect detected! Cease welding to avoid burn through". The warning 602 may be projected on the user's retina via the virtual retinal display and/or presented on a display of the vocational mask 130.

[0077] FIG. 7 illustrates an example of a method 700 for executing an artificial intelligence agent to determine certain information projected via a vocational mask of a user according to certain embodiments of this disclosure. The method 700 may be performed by processing logic that may include hardware (circuitry, dedicated logic, etc.), software, or a combination of both. The method 700 and/or each of their individual functions, subroutines, or operations may be performed by one or more processing devices of a computing device (e.g., any component (server 128, training engine 152, machine learning models 154, etc.) of cloud-based computing system 116, vocational mask 130, edge processor 132 (132.1, 132.2), peripheral haptic device 134, tool 136, and/or computing device 140 of FIG. 1) implementing the method 700. The method 700 may be implemented as computer instructions stored on a memory device and executable by the one or more processors. In certain implementations, the method 700 may be performed by a single processing thread. Alternatively, the method 700 may be performed by two or more processing threads, each thread implementing one or more individual functions, routines, subroutines, or operations of the methods.

[0078] For simplicity of explanation, the method 700 is depicted and described as a series of operations. However, operations in accordance with this disclosure can occur in various orders or concurrently, and with other operations not presented and described herein. For example, the operations depicted in the method 700 may occur in combination with any other operation of any other method disclosed herein. Furthermore, not all illustrated operations may be required to implement the method 700 in accordance with the disclosed subject matter. In addition, those skilled in the art will understand and appreciate that the method 700 could alternatively be represented as a series of interrelated states via a state diagram or events.

[0079] In some embodiments, one or more machine learning models may be generated and trained by the artificial intelligence engine and/or the training engine to perform one

or more of the operations of the methods described herein. For example, to perform the one or more operations, the processing device may execute the one or more machine learning models. In some embodiments, the one or more machine learning models may be iteratively retrained to select different features capable of enabling optimization of output. The features that may be modified may include a number of nodes included in each layer of the machine learning models, an objective function executed at each node, a number of layers, various weights associated with outputs of each node, and the like.

[0080] In some embodiments, a system may include the vocational mask **130**, which may include one or more virtual retinal displays, memory devices, processing devices, and other components as described herein. The processing devices may be communicatively coupled to the memory devices that store computer instructions, and the processing devices may execute the computer instructions to perform one or more of the steps of the method **700**. In some embodiments, the system may include a welding helmet and the vocational mask may be coupled to the welding helmet. In some embodiments, the vocational mask may be configured to operate across both visible light and high intensity ultraviolet light conditions. In some embodiments, the vocational mask may provide protection against welding flash. In some embodiments, the vocational mask may be integrated with goggles. In some embodiments, the vocational mask may be integrated with binoculars or a monocular.

[0081] At block **702**, the processing device may execute an artificial intelligence agent trained to perform at least one or more functions to determine certain information. The functions may include (i) identifying perception-based objects and features, (ii) determining cognition-based scenery to identify one or more material defects, one or more assembly defects, one or more acceptable features, or some combination thereof, and (iii) determining one or more recommendations, instructions, or both.

[0082] The artificial intelligence agent may include one or more machine learning models **154** trained to perform the functions. For example, at block **704**, one or more machine learning models **154** may be trained to (i) identify perception-based objects and features using training data that includes labeled inputs of images including certain objects and features mapped to labeled outputs of identities or characterizations of those objects and features. The machine learning models may be trained to analyze aspects of the objects and features to compare the aspects to known aspects associated with known objects and features, and the machine learning models may perceive the identity of the analyzed objects and features.

[0083] At block **706**, the one or more machine learning models **154** may be trained to (ii) determine cognition-based scenery to identify one or more material defects, one or more assembly defects, one or more acceptable features, or some combination thereof using training data that includes labeled input of scenery images of objects including material defects, assembly defects, and/or acceptable features mapped to labeled outputs that characterize and/or identify the material defects, assembly defects, and/or acceptable features. For example, one scenery image may include a portion of a submarine that includes parts that are welded together, and the machine learning models may be trained to cognitively analyze the scenery image to identify one or more portions of the scenery image that includes a welded

part with a material welding defect, a part assembly defect, and/or acceptable welded feature.

[0084] At block **708**, the one or more machine learning models **154** may be trained to (iii) determine one or more recommendations, instructions, or both using training data including labeled input of images (e.g., objects, products, tools, actions, etc.) and tasks to be performed (e.g., weld, grind, chamfer, debur, sand, polish, coat, etc.) mapped to labeled outputs including recommendations, instructions, or both. The processing device may provide (e.g., via the virtual retinal display, a speaker, etc.) images, video, and/or audio that points out the defects and provides instructions, drawings, and/or information pertaining to how to fix the defects.

[0085] In addition, the output from performing one of the functions (i), (ii), and/or (iii) may be used as input to the other functions to enable the machine learning models **154** to generate a combined output. For example, the machine learning models **154** may identify a defect (a gouge) and provide welding instructions on how to fix the defect by filling the gouge properly via the vocational mask **130**. Further, in some instances, the machine learning models **154** may identify several potential actions that the user can perform to complete the task and may aid the user's decision making by providing the actions in a ranked order of most preferred action to least preferred action or a ranked order of the action with the highest probability of success to the action with the lowest probability of success. In some embodiments, the machine learning models **154** may identify an acceptable feature (e.g., properly welded parts) and may output a recommendation to do nothing.

[0086] The artificial intelligence agent may further monitor one or more aspects of a virtual reality session. Based on the monitoring, the artificial intelligence agent may generate and provide, in real time, direction for performing a given task, during the virtual reality session, by a first user wearing the vocational mask. Alternatively, the artificial intelligence agent may generate suggested instructions by a master/teacher sharing the virtual reality session with an apprentice/student.

[0087] Embodiments are further contemplated in which the artificial intelligence agent may generate and provide, to various users of the shared virtual reality session, information in furtherance of completing a task where the various users are collaborators. In such collaborative environments, the various users of the shared virtual reality session may have different contexts, with such contexts being switchable among users. For example a context may be control of a machine that can be carried out locally or remotely, and these contexts may be exchanged among the different users in the shared virtual reality session. The artificial intelligence agent may adjust which user receives particular information based on such context switches.

[0088] At block **710**, the processing device may cause the certain information to be presented via the virtual retinal display. In some embodiments, the virtual retinal display may project an image onto at least one iris of the user to display alphanumeric data, graphic instructions, animated instructions, video instructions, or some combination thereof. In some embodiments, the vocational mask may include a stereo speaker to emit audio pertaining to the information. In some embodiments, the processing device may superposition the certain information on a display (e.g., virtual retinal display).

[0089] In some embodiments, the vocational mask may include a network interface configured to enable bidirectional communication with a second network interface of a second vocational mask. The bidirectional communication may enable transmission of real-time or near real-time audio and video data, recorded audio and video data, or some combination thereof. “Real-time” may refer to less than 2 seconds and “near real-time” may refer to between 2 and 20 seconds.

[0090] In some embodiments, in addition to the vocational mask, a system may include a peripheral haptic device. The vocational mask may include a haptic interface, and the haptic interface may be configured to perform bidirectional haptic sensing and stimulation using the peripheral haptic device and the bidirectional communication. The stimulation may include precise mimicking, vibration, and the like. For example, the stimulation may include performing mimicked gestures via the peripheral haptic device. In other words, a master user may be using a peripheral haptic device to perform a task and the gestures performed by the master user using the peripheral haptic device may be mimicked by the peripheral haptic device being used by an apprentice user. In such a way, the master user may train and/or guide the apprentice user how to properly perform a task (e.g., weld) using the peripheral haptic devices.

[0091] The haptic interface may be communicatively coupled to the processing device. The haptic interface may be configured to sense, from the peripheral haptic device, hand motions, texture, temperature, vibration, slipperiness, friction, wetness, pulsation, stiction, friction, and the like. For example, the haptic interface may detect keystrokes when a user uses a virtual keyboard presented via the vocational mask using a display (e.g., virtual retinal display).

[0092] Further, the bidirectional communication provided by the vocational mask(s) and/or computing devices may enable a master user of a vocational mask and/or computing device to view and/or listen to the real-time or near real-time audio and video data, recorded audio and video data, or some combination thereof, and to provide instructions to the user via the vocational mask being worn by the user. In some embodiments, the bidirectional communication provided by the vocational mask(s) and/or computing devices may enable the user of a vocational mask and/or computing device to provide instructions to a set of students and/or apprentices via multiple vocational masks being worn by the students and/or apprentices. This technique may be beneficial for a teacher, collaborator, master user, and/or supervisor that is training the set of students.

[0093] In some embodiments, the user wearing a vocational mask may communicate with one or more users who are not wearing a vocational mask. For example, a teacher and/or collaborator may be using a computing device (e.g., smartphone) to see what a student is viewing and hear what the student is hearing using the bidirectional communication provided by the vocational mask worn by the student. The bidirectional communication provided by the vocational mask may enable a teacher or collaborator to receive, using a computing device, audio data, video data, haptic data, or some combination thereof, from the vocational mask being used by the user.

[0094] Additionally, the teacher and/or collaborator may receive haptic data, via the computing device, from the vocational mask worn by the student. The teacher and/or

collaborator may transmit instructions (e.g., audio, video, haptic, etc.), via the computing device, to the vocational mask to guide and/or teach the student how to perform the task (e.g., weld) in real-time or near real-time.

[0095] In another example, the bidirectional communication may enable a user wearing a vocational mask to provide instructions to a set of students via a set of computing devices (e.g., smartphones). In this example, the user may be a teacher or collaborator and may be teaching a class or lesson on how to perform a task (e.g., weld) while wearing the vocational mask.

[0096] In some embodiments, the vocational mask may include one or more sensors to provide information related to geographical position, pose of the user, rotational rate of the user, or some combination thereof. In some embodiments, a position sensor may be used to determine a location of the vocational mask, an object, a peripheral haptic device, a tool, etc. in a physical space. The position sensor may determine an absolute position in relation to an established reference point. In some embodiments, the processing device may perform physical registration of the vocational mask, an object being worked on, a peripheral haptic device, a tool (e.g., welding gun, sander, grinder, etc.), etc. to map out the device in an environment (e.g., warehouse, room, underwater, etc.) in which the vocational mask, the object, the peripheral haptic device, etc. is located.

[0097] In some embodiments, the vocational mask may include one or more sensors including vocation imaging band specific cameras, visual band cameras, stereo microphones, acoustic sensors, or some combination thereof. The acoustic sensors may sense welding clues based on audio signatures associated with certain defects or issues, such as burn through. Machine learning models **154** may be trained using inputs of labeled audio signatures, labeled images, and/or labeled videos mapped to labeled outputs of defects. The artificial intelligence agent may process received sensor data, such as images, audio, video, haptics, etc., identify an issue (e.g., defect), and provide a recommendation (e.g., stop welding due to detected potential burn through) via the vocational mask.

[0098] In some embodiments, the vocational mask may include an optical bench that aligns the virtual retinal display to one or more eyes of the user.

[0099] In some embodiments, the processing device is configured to record the certain information, communications with other devices (e.g., vocational masks, computing devices), or both. The processing device may store certain information and/or communications as data in the memory device communicatively coupled to the processing device, and/or the processing device may transmit the certain information and/or communications as data feeds to the cloud-based computing system **116** for storage.

[0100] FIG. 8 illustrates an example of a method **800** for transmitting instructions for performing a task via bidirectional communication between a vocational mask and a computing device according to certain embodiments of this disclosure. The method **800** may be performed by processing logic that may include hardware (circuitry, dedicated logic, etc.), software, or a combination of both. The method **800** and/or each of their individual functions, subroutines, or operations may be performed by one or more processing devices of a computing device (e.g., any component (server **128**, training engine **152**, machine learning models **154**, etc.) of cloud-based computing system **116**, vocational mask **130**,

edge processor **132** (**132.1**, **132.2**), peripheral haptic device **134**, tool **136**, and/or computing device **140** of FIG. **1**) implementing the method **700**. The method **800** may be implemented as computer instructions stored on a memory device and executable by the one or more processors. In certain implementations, the method **800** may be performed by a single processing thread. Alternatively, the method **800** may be performed by two or more processing threads, each thread implementing one or more individual functions, routines, subroutines, or operations of the methods.

[**0101**] For simplicity of explanation, the method **800** is depicted and described as a series of operations. However, operations in accordance with this disclosure can occur in various orders or concurrently, and with other operations not presented and described herein. For example, the operations depicted in the method **800** may occur in combination with any other operation of any other method disclosed herein. Furthermore, not all illustrated operations may be required to implement the method **800** in accordance with the disclosed subject matter. In addition, those skilled in the art will understand and appreciate that the method **800** could alternatively be represented as a series of interrelated states via a state diagram or events.

[**0102**] In some embodiments, one or more machine learning models may be generated and trained by the artificial intelligence engine and/or the training engine to perform one or more of the operations of the methods described herein. For example, to perform the one or more operations, the processing device may execute the one or more machine learning models. In some embodiments, the one or more machine learning models may be iteratively retrained to select different features capable of enabling optimization of output. The features that may be modified may include a number of nodes included in each layer of the machine learning models, an objective function executed at each node, a number of layers, various weights associated with outputs of each node, and the like.

[**0103**] At block **802**, while a first user wears a vocational mask **130** to perform a task, the processing device may receive, at one or more processing devices of the vocational mask **130**, one or more first data feeds from one or more cameras of the vocational mask **130**, sensors of the vocational mask **130**, peripheral haptic devices associated with the vocational mask **130**, microphones of the vocational mask **130**, or some combination thereof. In some embodiments, the vocational mask **130** may be attached to or integrated with a welding helmet and the task may be welding. In some embodiments, the task may be sanding, grinding, polishing, deburring, chamfering, coating, etc. The vocational mask **130** may be attached to or integrated with a helmet, a hat, goggles, binoculars, a monocular, or the like.

[**0104**] In some embodiments, the one or more first data feeds may include information related to video, images, audio, hand motions, haptics, texture, temperature, vibration, slipperiness, friction, wetness, pulsation, or some combination thereof. In some embodiments, the one or more first data feeds may include geographical position of the vocational mask **130**, and the processing device may map, based on the geographical position, the vocational mask **130** in an environment or a physical space in which the vocational mask **130** is located.

[**0105**] At block **804**, the processing device may transmit, via one or more network interfaces of the vocational mask **130**, the one or more first data feeds to one or more

processing devices of the computing device **140** of a second user. In some embodiments, the computing device **140** of the second user may include one or more vocational masks, one or more smartphones, one or more tablets, one or more laptop computers, one or more desktop computers, one or more servers, or some combination thereof. The computing device **140** may be separate from the vocational mask **130**, and the one or more first data feeds are at least one of presented via a display of the computing device **140**, emitted by an audio device of the computing device **140**, or produced or reproduced via a peripheral haptic device coupled to the computing device **140**. In some embodiments, the first user may be an apprentice, student, trainee, or the like, and the second user may be a master user, a trainer, a teacher, a collaborator, a supervisor, or the like.

[**0106**] At block **806**, the processing device may receive, from the computing device, one or more second data feeds pertaining to at least instructions for performing the task. The one or more second data feeds are received by the one or more processing devices of the vocational mask **130**, and the one or more second data feeds are at least one of presented via a virtual retinal display of the vocational mask **130**, emitted by an audio device (e.g., speaker) of the vocational mask **130**, or produced or reproduced via a peripheral haptic device **134** coupled to the vocational mask **130**.

[**0107**] In some embodiments, the instructions are presented, by the virtual retinal display of the vocational mask **130**, via augmented reality. In some embodiments, the instructions are presented, by the virtual retinal display of the vocational mask, via virtual reality during a simulation associated with the task. In some embodiments, the processing device may cause the virtual retinal display to project an image onto at least one iris of the first user to display alphanumeric data associated with the instructions, graphics associated with the instructions, animations associated with the instructions, video associated with the instructions, or some combination thereof.

[**0108**] At block **808**, the processing device may store, via one or more memory devices communicatively coupled to the one or more processing devices of the vocational mask **130**, the one or more first data feeds and/or the one or more second data feeds.

[**0109**] In some embodiments, the processing device may cause the peripheral haptic device **134** to vibrate based on the instructions received from the computing device **140**.

[**0110**] In some embodiments, the processing device may execute an artificial intelligence agent trained to perform at least one or more functions to determine certain information. The one or more functions may include (i) identifying perception-based objects and features, (ii) determining cognition-based scenery to identify one or more material defects, one or more assembly defects, one or more acceptable features, or some combination thereof, and (iii) determining one or more recommendations, instructions, or both.

[**0111**] FIG. **9** illustrates an example of a method **900** for implementing instructions for performing a task using a peripheral haptic device according to certain embodiments of this disclosure. The method **900** may be performed by processing logic that may include hardware (circuitry, dedicated logic, etc.), software, or a combination of both. The method **900** and/or each of their individual functions, subroutines, or operations may be performed by one or more processing devices of a computing device (e.g., any com-

ponent (server 128, training engine 152, machine learning models 154, etc.) of cloud-based computing system 116, vocational mask 130, edge processor 132 (132.1, 132.2), peripheral haptic device 134, tool 136, and/or computing device 140 of FIG. 1) implementing the method 900. The method 900 may be implemented as computer instructions stored on a memory device and executable by the one or more processors. In certain implementations, the method 900 may be performed by a single processing thread. Alternatively, the method 900 may be performed by two or more processing threads, each thread implementing one or more individual functions, routines, subroutines, or operations of the methods.

[0112] For simplicity of explanation, the method 900 is depicted and described as a series of operations. However, operations in accordance with this disclosure can occur in various orders or concurrently, and with other operations not presented and described herein. For example, the operations depicted in the method 900 may occur in combination with any other operation of any other method disclosed herein. Furthermore, not all illustrated operations may be required to implement the method 900 in accordance with the disclosed subject matter. In addition, those skilled in the art will understand and appreciate that the method 900 could alternatively be represented as a series of interrelated states via a state diagram or events.

[0113] In some embodiments, one or more machine learning models may be generated and trained by the artificial intelligence engine and/or the training engine to perform one or more of the operations of the methods described herein. For example, to perform the one or more operations, the processing device may execute the one or more machine learning models. In some embodiments, the one or more machine learning models may be iteratively retrained to select different features capable of enabling optimization of output. The features that may be modified may include a number of nodes included in each layer of the machine learning models, an objective function executed at each node, a number of layers, various weights associated with outputs of each node, and the like.

[0114] At block 902, the processing device may receive, at one or more processing devices of a vocational mask 130, first data pertaining to instructions for performing a task using a tool 136. The first data may be received from a computing device 140 separate from the vocational mask 130. In some embodiments, the computing device may include one or more peripheral haptic devices, one or more vocational masks, one or more smartphones, one or more tablets, one or more laptop computers, one or more desktop computers, one or more servers, or some combination thereof. In some embodiments, the task includes welding and the tool 136 is a welding gun.

[0115] At block 904, the processing device may transmit, via a haptic interface communicatively coupled to the one or more processing devices of the vocational mask 130, the first data to one or more peripheral haptic devices 134 associated with the tool 136 to cause the one or more peripheral haptic devices 134 to implement the instructions by at least vibrating in accordance with the instructions to guide a user to perform the task using the tool 136.

[0116] At block 906, responsive to the one or more peripheral haptic devices 134 implementing the instructions, the processing device may receive, from a haptic interface, feedback data pertaining to one or more gestures, motions,

surfaces, temperatures, or some combination thereof. The feedback data may be received from the one or more peripheral haptic devices 134, and the feedback data may include information pertaining to the user's compliance with the instructions for performing the task.

[0117] At block 908, the processing device may transmit, to the computing device 140, the feedback data. In some embodiments, transmitting the feedback data may cause the computing device 140 to produce an indication of whether the user complied with the instructions for performing the task. The indication may be produced or generated via a display, a speaker, a different peripheral haptic device, or some combination thereof.

[0118] In some embodiments, in addition to the first data being received, video data may be received at the processing device of the vocational mask 130, and the video data may include video pertaining to the instructions for performing the task using the tool 136. In some embodiments, the processing device may display, via a virtual retinal display of the vocational mask 130, the video data. In some embodiments, the video data may be displayed concurrently with the instructions being implemented by the one or more peripheral haptic devices 134.

[0119] In some embodiments, in addition to the first data and/or video data being received, audio data may be received at the processing device of the vocational mask 130, and the audio data may include audio pertaining to the instructions for performing the task using the tool 136. In some embodiments, the processing device may emit, via a speaker of the vocational mask 130, the audio data. In some embodiments, the audio data may be emitted concurrently with the instructions being by the one or more peripheral haptic devices 134 and/or with the video data being displayed by the virtual retinal display. That is, one or more of video, audio, and/or haptic data pertaining to the instructions may be used concurrently to guide or instruct a user how to perform a task.

[0120] In some embodiments, in addition to the first data, video data, and/or audio data being received, virtual reality data may be received at the processing device of the vocational mask 130, and the virtual reality data may include virtual reality multimedia representing a simulation of a task. The processing device may execute, via at least a display of the vocational mask 130, playback of the virtual reality multimedia. For example, an artificial intelligent simulation generator may be configured to generate a virtual reality simulation for performing a task, such as welding an object using a welding gun. The virtual reality simulation may take into consideration various attributes, characteristics, parameters, and the like of the welding scenario, such as type of object being welded, type of welding, current amperage, length of arc, angle, manipulation, speed, and the like. The virtual reality simulation may be generated as multimedia that is presented via the vocational mask to a user to enable a user to practice, visualize, and experience performing certain welding tasks without actually welding anything.

[0121] In some embodiments, in addition to the first data, video data, audio data, and/or virtual reality data being received, augmented reality data may be received at the processing device of the vocational mask 130, and the augmented reality data may include augmented reality multimedia representing at least the instructions (e.g., via text, graphics, images, video, animation, audio). The processing

device may execute, via at least a display of the vocational mask **130**, playback of the augmented reality multimedia.

[0122] In some embodiments, the processing device may execute an artificial intelligence agent trained to perform at least one or more functions to determine certain information. The one or more functions may include (i) identifying perception-based objects and features, (ii) determining cognition-based scenery to identify one or more material defects, one or more assembly defects, one or more acceptable features, or some combination thereof, and/or (iii) determining one or more recommendations, instructions, or both. In some embodiments, the processing device may display, via a display (e.g., virtual retinal display or other display), the objects and features, the one or more material defects, the one or more assembly defects, the one or more acceptable features, the one or more recommendations, the instructions, or some combination thereof.

[0123] FIG. 10 illustrates an example computer system **1000**, which can perform any one or more of the methods described herein. In one example, computer system **1000** may include one or more components that correspond to the vocational mask **130**, the computing device **140**, the peripheral haptic device **134**, the tool **136**, one or more servers **128** of the cloud-based computing system **116**, or one or more training engines **152** of the cloud-based computing system **116** of FIG. 1. The computer system **1000** may be connected (e.g., networked) to other computer systems in a LAN, an intranet, an extranet, or the Internet. The computer system **1000** may operate in the capacity of a server in a client-server network environment. The computer system **1000** may be a personal computer (PC), a tablet computer, a laptop, a wearable (e.g., wristband), a set-top box (STB), a personal Digital Assistant (PDA), a smartphone, a smart-watch, a camera, a video camera, or any device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that device. Further, while only a single computer system is illustrated, the term “computer” shall also be taken to include any collection of computers that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methods discussed herein.

[0124] The computer system **1000** includes a processing device **1002**, a main memory **1004** (e.g., read-only memory (ROM), solid state drive (SSD), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM)), a static memory **1006** (e.g., solid state drive (SSD), flash memory, static random access memory (SRAM)), and a data storage device **1008**, which communicate with each other via a bus **1010**.

[0125] Processing device **1002** represents one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. More particularly, the processing device **1002** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets or processors implementing a combination of instruction sets. The processing device **1002** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device

1002 is configured to execute instructions for performing any of the operations and steps of any of the methods discussed herein.

[0126] The computer system **1000** may further include a network interface device **1012**. The computer system **1000** also may include a video display **1014** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), one or more input devices **1016** (e.g., a keyboard and/or a mouse), and one or more speakers **1018** (e.g., a speaker). In one illustrative example, the video display **1014** and the input device(s) **1016** may be combined into a single component or device (e.g., an LCD touch screen).

[0127] The data storage device **1008** may include a computer-readable medium **1020** on which the instructions **1022** embodying any one or more of the methodologies or functions described herein are stored. The instructions **1022** may also reside, completely or at least partially, within the main memory **1004** and/or within the processing device **1002** during execution thereof by the computer system **1000**. As such, the main memory **1004** and the processing device **1002** also constitute computer-readable media. The instructions **1022** may further be transmitted or received over a network **20** via the network interface device **1012**.

[0128] While the computer-readable storage medium **1020** is shown in the illustrative examples to be a single medium, the term “computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “computer-readable storage medium” shall also be taken to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0129] In some examples, systems and methods of the present disclosure are configured to implement a virtual coach or avatar to provide instructions, encouragement, etc. to the user. For example, a user interface may be configured to display and project the virtual coach via the virtual retinal display of a vocational mask as described herein.

[0130] FIGS. 11A-11C show an example user interface **1100** configured to display a virtual avatar or coach **1104**. The user interface **1100** presents an actual environment and objects (e.g., a tool **1108**, a workpiece **1112**, etc.) being viewed by the user through the vocational mask (e.g., through an eye opening, window, slot, etc. of the vocational mask **130**) as well as projected graphical elements including, but not limited to, the virtual coach **1104**, instructions **1116** pertaining to performing a task, etc. As described herein, the instructions **1116** may be generated by one or more machine learning models, or may be provided via one or more computing devices and/or other vocational masks being used by other users (e.g., master user, collaborator, teacher, supervisor, etc.). In some examples, the instructions **1116** and/or other commands or feedback, control signals, data, etc. may be provided to the user interface **1100** (e.g., to a computing device implementing the user interface **1100** at the vocational mask, such as the computing device **140**) via a cloud-based computing system **1120**.

[0131] The virtual coach **1104** is selected, generated, displayed, controlled, etc. based on one or more criteria as

described below in more detail. For example, the virtual coach **1104** may have one or more personas (e.g., different professions, costumes or uniforms, characters, historical figures, etc., non-human personas such as animals, etc.), demeanors, appearances, voices, or other characteristics. Characteristics of the virtual coach **1104** may be modified based on user feedback and/or user performance (e.g., in real-time, based on historical performance over time, etc.). The virtual coach **1104** provides instructions via the user interface **1100** of the vocational mask as the user performs a work task (e.g., a welding task). For example, the instructions may be provided as text instructions (e.g., via a text bubble including the instructions **1116**), voice instructions (e.g., audio instructions provided via speakers of the vocational mask as described herein), or combinations thereof.

[0132] Criteria for generating/controlling the virtual coach **1104** may include user characteristics, performance-based characteristics, task-based characteristics, or combinations thereof. For example, user characteristics may include, but are not limited to, user demographics or psychographics data, task-related physical characteristics (e.g., height, right-or-left-handedness, etc.), and so on. Performance-based characteristics may include, but are not limited to, historical performance data for the user and/or other users performing the same or similar task, real-time performance data collected as the user performs the current task, predictive data (e.g., generated by an ML model based on the historical performance data, the real-time performance data, etc.) indicative of predicted performance for a remainder of the current task, etc. Task-based characteristics correspond to characteristics of the specific task being performed and may include, but are not limited to, a type or category of the task (e.g., welding), an expected or average duration of the task (e.g., based on historical data), an amount of time the user has been performing the current task, an estimated amount of time remaining to perform the current task, a difficulty rating assigned to the task (e.g., on a numerical or other scale, such as a 1-10 scale), a number of times the user performed has performed the current task (e.g., in previous sessions), etc.

[0133] In some examples, the virtual coach **1104** is displayed in a fixed or static region of the user interface **1100**. In other examples, a position of the virtual coach **1104** in the user interface **1100** is variable or dynamic. For example, the virtual coach **1104** may be displayed different regions of the user interface **1100** based on the task being performed, movement of the user, movement of the tool **1108**, movement of the workpiece **1112**, etc. As shown in FIG. 11A, the virtual coach **1104** is displayed in an unoccupied region of the user interface **1100** (e.g., a region of the user interface **1100** that does not include the tool **1108**, the workpiece **1112**, etc. As the user moves (e.g., causes the vocational mask to move, which in turn causes a field-of-view of the vocational mask to move), the virtual coach **1104** may move to a different, unoccupied region of the user interface **1100**.

[0134] For example, the computing device may be configured to determine whether a current location of the virtual coach **1104** overlaps with other objects/graphical elements (e.g., occupies a same region as another object in an x-y coordinate system of the user interface **1100**), identify an unoccupied region of the user interface **1100**, and change the displayed location of the virtual coach **1104** to the unoccupied region. Similarly, a size of the virtual coach **1104** may be adjusted to fit within a particular unoccupied region of the

user interface **1100**. As shown in FIG. 11B, the workpiece **1112** substantially almost an entire field-of-view of the user interface **1100**. Accordingly, in this example, the virtual coach **1104** may be omitted. Conversely, as shown in FIG. 11C, the workpiece **1112** may not be in the field-of-view of the user interface **1100**. In this example, the virtual coach **1104** is enlarged and displayed in a central region of the user interface **1100**.

[0135] In some examples, the virtual coach **1104** may be presented performing a same or similar task as the user (e.g., performing a welding task on a workpiece similar to the workpiece **1112**, performing a welding task on a same portion of the workpiece **1112**, welding along a same or similar path or welding line, etc.).

[0136] In various examples, movement/actions of the virtual coach **1104** may be automatically controlled by the computing device (e.g., AI- or ML model-controlled) and/or controlled/responsive to commands from a trainer, teacher, or other user as described herein. For example, when controlled by the computing device, the virtual coach **1104** may perform preset/predetermined movements or actions (e.g., a repeated movement, such as a looped animation, corresponding to a welding task being performed, which may demonstrate preferred motion or movement, speed, posture, etc. for the welding task). Conversely, when responsive to a trainer, the virtual coach **1104** may perform one of a plurality of possible movements or actions in response to commands from the trainer.

[0137] In still other examples, in some operating modes, the virtual coach **1104** may be controlled based on actual movements/actions of a trainer. For example, the trainer may selectively “inhabit” the virtual coach **1104** (e.g., from a remote location). In an example, the trainer may be presented with, on a display, a duplicate of the user interface **1104** to observe the actions of the user while performing the task. The trainer may be prompted to initiate or selectively initiate a demonstration mode in which the movement of the virtual coach **1104** is controlled by the movement of the trainer. For example, the trainer may perform a same or similar welding task on a workpiece or simply mimic preferred movements or postures for the welding task, which are then presented to the user in the user interface **1100** via movements of the virtual coach **1104**. As an example, movement of the trainer may be monitored using various sensors as described herein, transmitted to the computing device associated with the vocational mask (e.g., via the cloud-based computing system **1120**), and presented to the user via the virtual coach **1104**.

[0138] In some examples, the appearance of the virtual coach **1104** may change based on whether the virtual coach **1104** is being controlled by the computing system or the trainer. For example, in response to the trainer taking control or inhabiting the virtual coach **1104**, the appearance of the virtual coach **1104** may change to a persona associated with the trainer.

[0139] In some examples, one or more display-related characteristics of the virtual coach **1104** may vary based on performance of the task by the user. For example, to reduce a likelihood of distracting the user while performing a welding task, a determination of whether to display the virtual coach **1104**, a size of the virtual coach **1104**, movement of the virtual coach, etc. may be triggered based on one or inputs corresponding to actions performed by the user (e.g., based on whether the user is actively welding, as

indicated by a welding button or trigger **1124** on the tool **1108** being actuated). For example, in response to the trigger **1124** being actuated, the virtual coach **1104** may be removed from the user interface **1100**, a size of the virtual coach **1104** may be decreased, movement of the virtual coach **1104** may be stopped, volume of audio instructions may be decreased (or, in some examples, increased), etc.

[0140] Additionally or alternatively, a graphical element representing the user, such as a virtual avatar or user **1128**, may be displayed via the user interface **1100**. For example, the virtual user **1128** may mirror/imitate a posture, movements, etc. of the user. In some examples, the virtual coach **1104** may overlay the virtual user **1128** (e.g., the virtual coach **1104** may be at least partially transparent such that the virtual user **1128** is visible through the virtual coach **1104**). In these examples, the virtual coach **1104** may be controlled to demonstrate a target or desired posture or movement. For example, the virtual coach **1104** may be represented as a looped or repeated animation demonstrating the task being performed by the user. Conversely a state or frame of animation of the virtual user **1128** may correspond to a current position or state of the actual user performing the welding task. According, the user can view a comparison between the user's movements/posture and the target or desired movements/posture.

[0141] Various other displayed characteristics of the virtual coach **1104** (and/or the virtual user **1128**) may change during performance of the task by the user. For example, welding tasks are typically associated with challenges including, but are not limited to, user coordination (e.g., coordination of wire feeding, hand position, tool alignment, straightness of a weld, etc.), heat control (e.g., sufficient heat to perform the weld but maintaining heat below an amount that could damage the workpiece **1112**), travel speed (e.g., travel speed of the tool **1108**), and various safety considerations (heat, presence of fumes/gases, etc.). Accordingly, in some examples, characteristics of the virtual coach **1104** may change based on performance data indicative of one or more of user coordination, heat control, travel speed, and safety considerations. Other performance data may include, but is not limited to, data indicative of an amount of welding material used, thickness of a weld, porosity of a weld, penetration of a weld, and so on.

[0142] In one example, a displayed characteristic such as a color of clothing or other element, persona, expression or demeanor, size, position on the user interface **1100**, etc. of the virtual coach **1104** may change based on the performance data. As one example, a color, persona, expression, etc. of the virtual coach **1104** may have two or more levels or tiers, such as a one level indicating satisfactory performance (e.g. measured indicator below a threshold) and another level indicating unsatisfactory performance (e.g., a measured indicator greater than or equal to the threshold). In an example, the displayed characteristic may include a first level indicating that the measured indicator is at least a first amount less than the threshold, a second level indicating that the measured indicator is approaching the threshold (e.g., within a certain range of the threshold), and a third level indicating that the measured indicator is greater than or equal to the threshold. The various levels may be represented by discrete states of the virtual coach **1104** and/or as a continuum or gradient.

[0143] For example, for user coordination, the measured indicator may correspond to a distance or amount of varia-

tion of an actual weld line from a target weld line and the threshold may correspond to a maximum allowable distance from the target weld line. For heat control, the measured indicator may correspond to a measured, sensed, estimated, or modeled heat value (e.g., based on inputs from sensors in the vocational mask, the tool **1108**, in a vicinity of the workpiece **1112**, etc.) and the threshold may include a first threshold corresponding to a minimum allowable heat and a second threshold corresponding to a maximum allowable heat. For travel speed, the measured indicator may correspond to a measured, sensed, estimated, or modeled speed (e.g., based on inputs from a sensor in the tool **1108** and/or attached to the user, a calculation based on observed movement of the tool **1108** within the user interface **1100**, etc.) and the threshold may include a first threshold corresponding to a minimum allowable travel speed and a second threshold corresponding to a maximum allowable travel speed. For the various safety considerations, the measured indicator may correspond to measured, sensed, estimated, or modeled values of environmental indicators such as heat, gases, etc. and the threshold may correspond to a maximum measured amount or value.

[0144] In one example, a color of the virtual coach **1104** (e.g., a color of the entire graphical element corresponding to the virtual coach **1104**, a color of clothing of the virtual coach **1104**, etc.) may cycle through three possible colors (e.g., green, yellow, and red) in accordance with the first, second, and third levels as described above. In another example, an expression or demeanor of the virtual coach **1104** may cycle through three possible expressions, such as a happy or smiling expression, a neutral or concerned expression, and an angry or fearful expression. In another example, a posture of the virtual coach **1104** may cycle through three possible postures, such as a positive posture (e.g., a "thumbs up" gesture), a neutral or concerned posture (e.g., a wagging finger), and a negative posture (e.g., a "thumbs down" gesture).

[0145] In some examples, the changing or cycling may be the same for each display characteristic (e.g., the same possible colors for each level of heat control, travel speed, etc.) or may be different for each display characteristic. As one example, as shown in FIG. 11C, the cycling between levels may be accompanied by one or more icons **1132** (e.g., a fire extinguisher for heat control) or other graphical elements indicating which performance data is being indicated by the cycling display characteristic. As another example, each display characteristic may be associated with a particular type of performance data. For example, cycling expression may indicate different coordination levels, cycling color may indicate different heat control levels, cycling posture may indicate different safety consideration levels, etc. In this manner, various display characteristics of the virtual coach **1104** may dynamically change to provide different types of information to the user corresponding to different performance or other types of indicators. As used herein, in some contexts "display characteristics" may refer to audio characteristics, such as characteristics of a voice or vocal instructions associated with the virtual coach **1104**, such as volume, tone, etc. of the voice.

[0146] Display characteristics of the virtual coach **1104** may change based on progress (e.g., a percentage or other measurement of a portion of the task that has been completed) and/or performance (e.g., one or more measures of performance, such as a calculated performance score or

grade). For example, performance may be measured based on a comparison between performance of the current task and a baseline, reference, or target performance level of the task, a comparison between the performance of the current task and previous performances of the task by the same user or other users, etc. In an example, performance may be calculated (e.g., as a performance score) based on one or more performance values, which may correspond to measured or sensed values, calculated values, modeled or estimated values, or combinations thereof. As one example, a performance score may be based on a calculation of an overall amount of variation of performance values from one or more target values (e.g., a running average of performance values), an amount of times respective performance values varied (e.g., by at least a predetermined amount) from one or more target values, etc.

[0147] The performance values may be obtained using one or more techniques, including, but not limited to, imaging techniques (e.g., image processing of images displayed via the user interface **1100** to monitor and measure characteristics of the weld and work environment), acoustic sensing techniques (e.g., using one or more acoustic sensors as described herein), modeling techniques (e.g., techniques responsive to inputs from various sensors in the work environment, vocational mask, on or around the user, etc.), AI and ML techniques, and so on.

[0148] In an example, the performance score is calculated (e.g., using one or more of the ML models **154**) based on a combination of performance values each corresponding to one or more of a plurality of measured, sensed, calculated, estimated, etc. values. For example, the performance score may be a numerical value, such as an integer or non-integer value (e.g., a value on a scale from 1 to 10, from 1 to 100, etc.) or a percentage. Each individual performance value may be calculated in accordance with a variation from a target value, where a maximum performance value (e.g., 100) is assigned for measurements within a minimum threshold of a target or baseline value and a minimum performance value (e.g., 0) is assigned for measurements greater than or equal to a maximum threshold of the target or baseline value. The target or baseline value may be a predetermined value (e.g., a predetermined target temperature), a dynamic value based on historical performance data (e.g., an average or other combination of temperatures measured during previous performances of the current task by the same or different users), or combinations thereof.

[0149] In an example, the performance score is an average or weighted average of the individual performance values. For example, one or more of the ML models **154** may be trained to obtain weights for each of the performance values based on assessments of overall weld quality (i.e., for a completed weld by the user and/or other users) and the various performance values calculated during performance of the respective welds. In other words, an ML model **154** may be trained to identify which performance values have the greatest effect on overall weld quality and assign weights accordingly. The weights may correspond to percentages or decimal values between 0 and 1, a ranking of the performance values (e.g., from 1 to n or from n to 1, where n is an integer greater than one), etc.

[0150] In this manner, the performance score may be continuously calculated and updated as the user performs the welding task and the virtual coach **1104** is generated and controlled based on the performance score (e.g., display

characteristics such as colors, persona, demeanor, etc. of the virtual coach **1104** are changed based on the performance score). The display characteristics may be changed between discrete values or levels as described herein, changed in a gradient manner, etc.

[0151] In some examples, the virtual coach **1104** (e.g., display characteristics of the virtual coach **1104**) may be controlled based in part on improvement or a rate of improvement (over time for the current task, a plurality of tasks, etc.) for the user and/or other users. For example, improvement may be measured based on improvement in the overall performance score and/or individual performance values for the current task, improvement from task-to-task, over time, for a plurality of tasks, etc. More specifically, various rates of improvement may be indexed to or correlated with (e.g., via an ML model) respective display characteristics and levels as described herein such that responsiveness to various display characteristics, for a given user or group of users, can be calculated. In other words, different users may show increased or decreased rates of improvement in response to different display characteristics of the virtual coach **1104** during specific stages or actions of a task. As one example, some users or types of users may be more responsive to higher levels (e.g., the third level described herein) and demonstrate greater rates of improvement while other users or types of users may be less responsive to higher levels and demonstrate lower rates of improvement or degraded performance. As described above, data indicative of improvement and/or rate of improvement may be referred to as “improvement data.”

[0152] Accordingly, the ML models according to the present disclosure may be trained to cycle through the various levels, omit or skip certain levels, use or not use certain colors or expressions, and change the display based on observed increases or decreases in performance, rates of improvement, etc. of various users over time. For example, training data may include labeled inputs (e.g., increased or decreased performance scores, rates of improvement, etc. of users over time) mapped to labeled outputs (e.g., various levels, colors or expressions, etc. to modify a display). The training data may be used to train one or more of the ML models.

[0153] FIG. 12 illustrates steps of an example method **1200** of controlling a user interface and virtual coach according to embodiments of the present disclosure. One or more computing systems, computing devices, processors or processing devices, AI/ML models, etc. as described herein may be configured to perform various steps of the method **1200**. While the method **1200** may be implemented for performing different types of tasks, the steps of the method **1200** are described with respect to a welding task for illustration purposes only.

[0154] At block **1204**, a user begins performing a welding task (e.g., on a workpiece.). Beginning to perform the welding task may include, but is not limited to, putting on a vocational mask as described herein and powering up and initializing welding equipment. In some examples, the user may log in or otherwise provide identification data (e.g., to a computing system) that includes or provides an indication of various user characteristics, previous performance data (e.g., historical data), user settings or preferences, etc. In some examples, beginning to perform the welding task may include providing an indication of the welding task, such as a type of welding task, a type of workpiece, etc. In some

examples, the method **1200** includes, at block **1204**, obtaining data associated with the user and/or the welding task (e.g., from the database **129**). The data may include, is not limited to, data identifying one or more characteristics of the user, historical data associated with performance of the welding task or other task by the user or other users, operating characteristics (e.g., target measurement or performance values) associated with the welding task or work-piece, etc.

[0155] At block **1208**, the method **1200** includes generating a virtual coach (and/or, in some examples, a virtual user) and displaying the virtual coach on a user interface of the vocational mask. Initially, the virtual coach may be a default (i.e., predetermined) virtual coach for the user or group of users, a virtual coach selected based on characteristics of the user, a virtual coach selected based on previous performances (e.g., welding sessions) of the user, etc.

[0156] At block **1212**, the method **1200** includes obtaining performance data while the user performs the welding task. The performance data may include measured, sensed, calculated, estimated, and/or modeled values as described herein. At block **1216**, the method **1200** includes obtaining performance values based on the performance data. The performance values may be calculated based on comparisons between the performance values and one or more respective reference, baseline, or target values as described herein.

[0157] At block **1220**, the method **1200** includes generating a performance score based on the performance data and/or performance values. For example, generating the performance score may include providing the performance values and/or performance data, historical performance data (e.g., for the user and/or other users), user characteristics (e.g., for the user and/or other users), or any other data described herein to one or more ML models trained to generate the performance score. In an example, the performance score is a weighted average of the performance values, and the one or more ML models are configured to calculate and assign weights to the performance values as described herein.

[0158] At block **1224**, the method **1200** includes controlling the virtual coach (e.g., controlling one or more display characteristics of the virtual coach) based on the performance score and, in some examples, further based on historical performance scores, values, and/or data for the user and/or other users. For example, the display characteristics may simply be controlled based on a comparison between the performance score and various ranges of performance scores (e.g., a determination of whether the performance score falls within a first range, a second range, a third range, etc.). As one example, a first range may be defined as a range of performance scores between 0 and m , a second range may be defined as a range between m and n , and a third range may be defined as a range between n and 100. The ranges may be same or different sizes, the ranges may be variable, and more or fewer ranges may be implemented.

[0159] In other examples, the display characteristics may be controlled based on comparisons between respective performance values and various ranges of performance values (e.g., comparisons between temperature values and ranges, travel speed values and ranges, etc.). In other words, in various examples, the display characteristics may be controlled based on an overall performance score (e.g., a

performance score based on a combination of a plurality of performance values) and/or based on a plurality of different performance values corresponding to different measured (or calculated, modeled, etc.) values.

[0160] In other examples, controlling the virtual coach includes controlling the display characteristics further based on improvement data as described herein. For example, the method **1200** may include providing the performance score, performance data, etc. along with improvement data to an ML model trained to generate one or more virtual coach control outputs to control the display characteristics of the virtual coach. For example, rather than simply controlling the display characteristics based on the performance score and a comparison to various performance ranges as described above, the ML model may be configured to select/adjust various display characteristics further based on the improvement data. As one example, the performance ranges may be adjusted based on the improvement data (e.g., one or more performance ranges may be increased or decreased based on improvement data indicating that the user or other users or more likely or less likely to improve in response to specific display characteristics). As another example, one or more performance ranges may be eliminated. As another example, various display characteristics may be omitted (e.g., one or more colors, expressions, personas, etc. may not be displayed to a given user while other colors, expressions, personas, etc. are more likely to be displayed to the user). As another example, audio characteristics may be adjusted (e.g., volume may be increased or decreased, voice or audio tones may be omitted or favored, etc.).

[0161] Accordingly, rather than simply controlling the display characteristics in the same manner for all users (or in the same manner for a given user for all tasks), the ML model is configured to generate control outputs based on improvement data indicative of which display characteristics are most effective for improving performance of a user or group of users.

[0162] FIG. 13 illustrates steps of an example method **1300** of conducting a shared virtual reality session between an apprentice and a master. At block **1304**, method **1300** includes wearing, by a first user, a first vocational mask. The first vocational mask may include a virtual retinal display, a memory device storing instructions, and a processing device communicatively coupled to the memory and the virtual retinal display. The first vocational mask may correspond to any of the various embodiments of the same discussed above.

[0163] At block **1308**, the method includes executing, using the processing device, the instructions stored in the memory device, while at block **1313**, the method includes sharing, in response to execution of the instructions, a virtual reality session with a computing device associated with a second user. At block **1316**, the sharing of the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with the computing device associated with the second user, wherein the first user is an apprentice and the second user is a master. Through the shared virtual reality session, a master may conduct training of the apprentice in the performance of a specific task, such as welding, brazing, soldering, various mechanical tasks, and so on. Through the use of a network such as network **20** of FIG. 1, this training may be conducted

even when the master and the apprentice are at geographically disparate locations with respect to one another.

[0164] At block 1320, method 1300 includes presenting, to the first user using the first vocational mask, data received from the computing device associated with the second user. At block 1324, the data is received in a plurality of formats, and wherein the data includes directions for performing a task by the first user under direction from the second user.

[0165] In various embodiments, method 1300 may include projecting the data, using the first vocational mask, to a retina of the first user, wherein the data is presented in one or more of the plurality of formats. Sharing the virtual reality session with the computing device associated with a second user may include, in one non-limiting example, sharing the virtual reality session with a second vocational mask worn by the second user. In addition to a second vocational mask, the disclosure further contemplates, as the computing device associated with the second user, tablets, smartphones, laptop computers, desktop computers, and any other computing device suitable for sharing a virtual reality session with the first user via the first vocational mask.

[0166] The plurality of data formats may include one or more of the following image data, video data, audio data, and haptic feedback. For example, during a training session, a master may convey video data indicating a proper procedure for carrying out a particular task, still image data indicating an appearance of an end result of the task (e.g., a properly formed weld), and audio data to supplement the other formats. In a vocational mask that includes a haptic peripheral, data may be conveyed in the form of haptic feedback in order to instruct an apprentice of the proper feel for carrying out the particular task for which instruction is being provided.

[0167] The method may, in various embodiments, include sharing the virtual reality session when the first user and the second user being located remotely with respect to one another. Such remote locales may, in various examples, include different locations within the same facility wherein direct verbal communications may be impractical or impossible, in different cities, or in different geographic regions for which the communications are conducted using long-distance communication mechanisms (e.g., via the internet).

[0168] Method 1300 is not limited to a master instructing a single apprentice. Accordingly method 1300 may include, in various embodiments, sharing the virtual reality session among a plurality of computing devices associated with respective ones of a plurality of users, wherein the plurality of computing devices includes the first vocational mask and one or more additional vocational masks worn by respective ones of a subset of a plurality of users, wherein the subset of the plurality of users includes the first user and one or more additional users, wherein the one or more additional users are apprentices. In some instances, at least one apprentice may be remotely located with respect to at least one other apprentice, and may also be located remotely with respect to the master. As such, the method contemplates a single master conducting training with multiple apprentices, with the master and at least some of the multiple apprentices being remotely located with respect to one another. Thus, to use one example, a master in a first geographic location may, in one embodiment of the method, concurrently instruct a first apprentice in a second geographic location, a second apprentice in a third geographic location, and a third appren-

tice in fourth geographic location, wherein the first, second, third, and fourth geographic locations are all remote with respect to one another.

[0169] The disclosure further contemplates method embodiments in which the first user provides, using the first vocational mask, feedback to the computing device associated with the second user, wherein the feedback is provided in one or more of the plurality of formats.

[0170] In various method embodiments, executing the instructions stored in the memory device comprises executing an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time (e.g., less than 2 seconds), directions for performing the task to the first user during the virtual reality session. In one example, the task is welding, and wherein the method further comprises monitoring, using the artificial intelligence agent, one or more properties of a weld formed by the first user and adjusting, using the artificial intelligence agent, one or more properties of the weld based on the monitoring. The use of the artificial intelligence agent may be similarly applied to other tasks as well.

[0171] FIG. 14 illustrates examples of respective user interfaces for both an apprentice and a master during training to perform a particular task. The illustrated example includes an apprentice user interface (UI) 1401, and a master UI 1402. Through the apprentice UI, a work piece 1410 comprising a cylindrical object and a flat object are shown, along with a welding gun 1411. In the shared virtual reality session, the work piece 1410 along with the welding gun 1411 can be seen, allowing the master to monitor the performance of the task by the apprentice. The work piece 1410 and the welding gun 1411 may be virtual graphical objects presented concurrently by the UI 1401 and the master UI 1402.

[0172] In the upper portion of the apprentice UI, instructions for carrying out the task are displayed. As the example task here is welding, the instructions include adjusting a wire speed and voltage, as well as the positioning of the welder at 45° to weld a bevel around the top of the cylinder. The performance of the task is monitored by a master through the master UI 1402.

[0173] As discussed elsewhere herein, an AI agent may be executed on the vocational mask of the apprentice. The AI agent may provide monitoring in addition to that provided by the master through the master UI. In this example, the AI agent may monitor the weld joint between the cylinder and the flat object for quality and proper form. The AI agent may be trained, using training data, to identify images, screens, objects, and the like in the virtual reality session, determine contextual meaning corresponding to the identified images, screens, objects, etc., and determine, based on the contextual meaning, recommended directions and/or information to present to the apprentice. The training data may include labeled inputs (e.g., images, video, audio, haptic data, etc.) mapped to labeled outputs (e.g., directions, information, recommendations). As a result of the monitoring by the AI agent on the apprentice vocational mask, the master UI 1402 displays information that the weld joint is too thin and that the apprentice needs to increase the travel speed of the weld gun. As a result, or concurrently therewith, the apprentice UI 1401 displays an additional instruction that the apprentice is to increase the travel speed of the weld gun, along with the reasoning behind this instruction.

[0174] Generally speaking the disclosure contemplates a system comprising a first vocational mask configured to be worn by a first user, wherein the first vocational mask comprises: a virtual retinal display; a memory device storing instructions; and a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to: share a virtual reality session with a computing device associated with a second user, wherein the first vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master; and receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats from the second user, the data comprising directions for performing a task by the first user under direction from the second user.

[0175] In various embodiments, the first vocational mask is configured to receive the data from a second vocational mask worn by the second user.

[0176] The plurality of formats (by which data may be received at the first includes audio data. In various embodiments, the plurality of formats includes video data. In various embodiment, the plurality of formats includes image data.

[0177] In some embodiments, the first vocational mask includes one or more peripheral haptic devices, and wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to present, as one of the plurality of formats, haptic feedback to the first user.

[0178] The first vocational mask in various embodiments is configured to project data in at least one of the plurality of formats to a retina of the first user via the virtual retinal display.

[0179] The first vocational mask is configured to convey feedback data in one or more of the plurality of formats to the computing device of the second user.

[0180] In various embodiments, the first user is located remotely with respect to the second user, and thus the first vocational mask is located remotely with respect processing device used by the second user.

[0181] In some embodiments, the system further comprises a plurality of vocational masks including the first vocational mask, wherein each of the plurality of vocational masks is configured to conduct bidirectional communications with the computing device associated with the second user. In such embodiments, at least a subset of the plurality of vocational masks may be configured to receive and present data in a plurality of formats from the computing device associated with the second user, wherein respective users of the subset are apprentices receiving the directions for performing the task.

[0182] In various embodiments, the instructions, when executed by the processing device, are configured to execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session. In one example embodiment, the task is welding, and wherein the artificial intelligence agent is trained to monitor one or more properties of a weld formed by the first user, and further trained to adjust directions to the first user based on the one or more properties of the weld.

[0183] The disclosure further contemplates a non-transitory computer-readable medium storing instructions that, when executed by a processing device of a first computing system, cause the first computing system to generate, for a first user wearing a first vocational mask that comprises the first computing system, a virtual reality session, wherein the first vocational mask includes a virtual retinal display; share the virtual reality session with a second computing system associated with a second user, wherein sharing the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with a second computing system associated with the second user, wherein the first user is an apprentice and wherein the second user is a master; and present, to the first user, using the first vocational mask, data received from the computing device associated with the second user, wherein the data is received in a plurality of formats, and wherein the data comprises directions for performing a task by the first user under direction from the second user.

[0184] In various embodiments, the first vocational mask includes a virtual retinal display, and wherein the instructions are executable to cause data to be displayed in at least a subset of the plurality of formats as visual data displayed to the first user via the virtual retinal display. In various embodiments, the subset of the plurality of formats includes video data and image data.

[0185] Embodiments are contemplated in which the first vocational mask includes at least one haptic peripheral device, and wherein the instructions are executable to cause haptic feedback, as one of the plurality of formats, to be presented to the first user via the at least one haptic peripheral device.

[0186] The disclosure further contemplates that the instructions, when executed by the processing device of the first computing system, cause the first computing system to execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

[0187] The disclosure further contemplates that the instructions, when executed by the processing device of the first computing system, cause the first computing system to share the virtual reality session among a plurality of computing devices associated with respective ones of a plurality of users, wherein the plurality of computing devices includes the first vocational mask and one or more additional vocational masks worn by respective ones of a subset of a plurality of users, wherein the subset of the plurality of users includes the first user and one or more additional users, wherein the one or more additional users are apprentices.

[0188] The disclosure further contemplates that the instructions, when executed by the processing device of the first computing system, cause the first computing system to send feedback data to the second computing system.

[0189] The disclosure additionally contemplates a vocational mask configured to be worn by a first user, wherein the vocational mask comprises: a virtual retinal display; a memory device storing instructions; one or more peripheral haptic devices; a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the processing device executes the instructions to: share a virtual reality session with a computing device associated with a second user, wherein the vocational mask is configured to conduct bidirectional communications with

the computing device of the second user, wherein the first user is an apprentice and the second user is a master; receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats from the second user, the data comprising directions for performing a task by the first user under direction from the second user; cause the vocational mask to present, via at least one of the one or more peripheral haptic devices, haptic feedback to the first user; and execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

[0190] FIG. 15 illustrates steps of an example of a method of conducting a shared virtual reality session between a first user and a second user operating in switchable first and second contexts. Method 1500 includes, at block 1504, wearing, by a first user, a first vocational mask, the first vocational mask having a virtual retinal display, a memory device storing instructions, and a processing device communicatively coupled to the memory and the virtual retinal display. At block 1508, the method includes executing, using the processing device, the instructions stored in the memory device. At block 1515, the method includes sharing, in response to execution of the instructions, a virtual reality session with a computing device associated with a second user, wherein sharing the virtual reality session comprises conducting bidirectional communications with the computing device of the second user, wherein the first user is operating in a first context and wherein the second user is operating in a second context. At block 1516, the method includes presenting, via the virtual retinal display to the first user, data in selected ones of a plurality of formats received from the computing device associated with the second user, the data comprising information regarding a task to be performed, at least in part, by the first user.

[0191] In various embodiments, the method includes executing instructions to cause the first user to switch to the second context concurrent with the second user switching to the first context. In some of these embodiments, the first user is a teacher and the second user is a student. Embodiments are also possible and contemplated in which the first and second users are collaborators on the task.

[0192] Various embodiments of the method include comprising the first vocational mask receiving the data from a second vocational mask worn by the second user.

[0193] In some embodiments, the plurality of formats includes video data and still image data. Embodiments are further possible and contemplated in which the plurality of formats includes audio data. Some embodiments of the method may include executing instructions to cause the first vocational mask to present, as one of the plurality of formats, haptic feedback to the first user via a haptic peripheral device coupled to the first vocational mask.

[0194] In various embodiments, the first and second users are collaborators on the task, wherein the method further comprises the artificial intelligence agent providing a first description indicating a first portion of the task to be performed by the first user and a second portion of the task to be performed by the second user.

[0195] FIG. 16 illustrates examples of respective user interfaces for first and second users sharing a virtual reality session with switchable contexts. In the illustrated example, a first user UI (user interface) 1601 and a second user UI

1602 are shown. The first user UI 1601 may be a display from a vocational mask as discussed above. The second user UI 1602 may be that of a computing device used by the second user. The computing device associated with the second user may be another instance of a vocational mask in one implementation. However, the disclosure contemplates that the computing device associated with the second user may be a laptop computer, a desktop computer, a smart phone, a tablet computer, or any other suitable computing device.

[0196] The first and second users in the example shown may be collaborators working to perform a task together, even though they may otherwise be separated in distance. For example, the task carried out may be a repair on a complex piece of equipment that can be controlled locally in some instances and remotely in others. The repair may at various times utilize local control of the equipment, while utilizing remote control of the equipment at other times. For example, the repair operation may include the first user controlling the equipment locally, while the second user remotely performs software updates remotely.

[0197] The operation involving the first and second users may thus include separate contexts. Thus, in the example provided herein, one of the users may perform an active role during the repair operation, while the other user may perform an observational role. Furthermore, these contexts may be switched between the users during the performance of the tasks.

[0198] In the upper portion of the drawing, the first user UI 1601 is operating in a first context (Context #1) and the second user UI 1602 is operating in a second context (Context #2). The first context in this example is an operational context, and initially includes instructions for the first user to perform various steps of the task, while the second context is an observational context. After the first user has completed the outlined steps, the context may be switched, with the first (operational) context being transferred to second user UI 1602 while the second (observational) contexts being transferred to first user UI 1601. After this switch, the second user may perform portions of the task as outlined in the instruction displayed on second user UI 1602, while the first user may assume an observational role.

[0199] The example discussed herein is one in which the first and second users are collaborators in the performance of a task. However, the disclosure is not limited to such examples. Per the discussion above, the first and second users may have a master-apprentice or teacher-student relationship as well. The switching of contexts may enable a master/teacher to demonstrate a task with the apprentice/student observing, before switching contexts to allow the apprentice/student to perform the previously demonstrated task while the master/teacher observes. The disclosure further contemplates the use of the systems and methods described here in an augmented environment. For example, a surgeon may carry out a complex surgery on a patient with augmentation of another user in an observational context through which advice and instruction may be relayed. A doctor-resident relationship is also contemplated within the contexts discussed herein. Per various embodiments discussed above, the disclosure also contemplates examples in which multiple users operate in a first context with one or more additional users operating in a second context. Examples with multiple users and more than two contexts are also possible and contemplated.

[0200] Accordingly, the disclosure contemplates a system comprising a first vocational mask configured to be worn by a first user, wherein the first vocational mask comprises: a virtual retinal display; a memory device storing instructions; and a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to: share a virtual reality session with a computing device associated with a second user, wherein the first vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is operating in a first context and wherein the second user is operating in a second context; receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats received from the computing device associated with the second user, the data comprising information regarding a task to be performed, at least in part, by the first user.

[0201] In various embodiments, the processing device is further configured to execute instructions to cause the first user to switch to the second context concurrent with the second user switching to the first context.

[0202] In various embodiments, the first user is a teacher and the second user is a student.

[0203] In various embodiments, the first and second users are collaborators on the task.

[0204] In various embodiments, the first vocational mask is configured to receive the data from a second vocational mask worn by the second user.

[0205] In various embodiments, the plurality of formats includes video data and still image data.

[0206] In various embodiments, the plurality of formats includes audio data.

[0207] In various embodiments, the first vocational mask includes one or more peripheral haptic devices, and wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to present, as one of the plurality of formats, haptic feedback to the first user.

[0208] In various embodiments, the first vocational mask is configured to project data in at least one of the plurality of formats to a retina of the first user via the virtual retinal display.

[0209] In various embodiments, the first user is located remotely with respect to the second user.

[0210] In various embodiments, the instructions, when executed by the processing device, are configured to execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, information regarding the task to the first user during the virtual reality session. In such embodiments, the information regarding the task may comprise directions for performing the task. In some embodiments in which the first and second users are collaborators on the task, the information may comprise a first description indicating a first portion of the task to be performed by the first user, wherein the artificial intelligence agent is further configured to generate a second description indicating a second portion of the task to be performed by the second user.

[0211] The disclosure further contemplates a non-transitory computer-readable medium storing instructions that, when executed by a processing device of a first computing

system, cause the first computing system to: generate, for a first user wearing a first vocational mask that comprises the first computing system, a virtual reality session, wherein the first vocational mask includes a virtual retinal display; share, in response to execution of the instructions, a virtual reality session with a computing device associated with a second user, wherein sharing the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with a second computing system associated with the second user; operate, by the first user using the first vocational mask, in a first context concurrent with the second user operating in a second context using the second computing system; and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats received from the computing device associated with the second user, the data comprising information regarding a task to be performed, at least in part, by the first user.

[0212] In various embodiments, the instructions are further executable to cause the first user to switch to the second context concurrent with the second user switching to the first context.

[0213] In various embodiments, wherein the instructions are further executable to operate an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, information regarding the task to the first user during the virtual reality session. In such embodiments, the information regarding the task may comprise directions for performing the task. The first and second users may be collaborators on the task, and wherein the information may comprise a first description indicating a first portion of the task to be performed by the first user, wherein the instructions are further executable to cause the artificial intelligence agent to generate a second description indicating a second portion of the task to be performed by the second user.

[0214] In various embodiments, the first user is a teacher and the second user is a student. Embodiments are further possible and contemplated in which the first and second users are collaborators on the task.

[0215] The disclosure further contemplates a vocational mask configured to be worn by a first user, wherein the vocational mask comprises: a virtual retinal display; a memory device storing instructions; one or more peripheral haptic devices; a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the processing device executes the instructions to: generate, for a first user wearing a first vocational mask that comprises the first computing system, a virtual reality session, wherein the first vocational mask includes a virtual retinal display; share, in response to execution of the instructions, a virtual reality session with a computing device associated with a second user, wherein sharing the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with a second computing system associated with the second user; operate, by the first user using the first vocational mask, in a first context concurrent with the second user operating in a second context using the second computing system; and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats received from the computing device associated with the second user, the data comprising information regarding a task to be performed, at least in part, by the first user.

CLAUSES

[0216] 1. A system comprising:

[0217] a first vocational mask configured to be worn by a first user, wherein the first vocational mask comprises:

[0218] a virtual retinal display;

[0219] a memory device having instructions stored thereon; and

[0220] a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to:

[0221] share a virtual reality session with a computing device associated with a second user, wherein the first vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master; and

[0222] receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats from the second user, the data comprising directions for performing a task by the first user under direction from the second user.

[0223] 2. The system of any clause herein, wherein the first vocational mask is configured to receive the data from a second vocational mask worn by the second user.

[0224] 3. The system of any clause herein, wherein the plurality of formats includes audio data.

[0225] 4. The system of any clause herein, wherein the plurality of formats includes video data.

[0226] 5. The system of any clause herein, wherein the plurality of formats includes image data.

[0227] 6. The system of any clause herein, wherein the first vocational mask includes one or more peripheral haptic devices, and wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to present, as one of the plurality of formats, haptic feedback to the first user.

[0228] 7. The system of any clause herein, wherein the first vocational mask is configured to project data in at least one of the plurality of formats to a retina of the first user via the virtual retinal display.

[0229] 8. The system of any clause herein, wherein the first vocational mask is configured to convey feedback data in one or more of the plurality of formats to the computing device of the second user.

[0230] 9. The system of any clause herein, wherein the first user is located remotely with respect to the second user.

[0231] 10. The system of any clause herein, further comprising a plurality of vocational masks including the first vocational mask, wherein each of the plurality of vocational masks is configured to conduct bidirectional communications with the computing device associated with the second user.

[0232] 11. The system of any clause herein, wherein at least a subset of the plurality of vocational masks are configured to receive and present data in a plurality of formats from the computing device associated with the second user, wherein respective users of the subset are apprentices receiving the directions for performing the task.

[0233] 12. The system of any clause herein, wherein the instructions, when executed by the processing device, are configured to execute an artificial intelligence agent trained

to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

[0234] 13. The system of any clause herein, wherein the task is welding, and wherein the artificial intelligence agent is trained to monitor one or more properties of a weld formed by the first user, and further trained to adjust directions to the first user based on the one or more properties of the weld.

[0235] 14. A method comprising:

[0236] wearing, by a first user, a first vocational mask, the first vocational mask having a virtual retinal display, a memory device storing instructions, and a processing device communicatively coupled to the memory device and the virtual retinal display;

[0237] executing, using the processing device, the instructions stored in the memory device;

[0238] sharing, in response to execution of the instructions, a virtual reality session with a computing device associated with a second user, wherein sharing the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with the computing device associated with the second user, wherein the first user is an apprentice and the second user is a master; and

[0239] presenting, to the first user using the first vocational mask, data received from the computing device associated with the second user, wherein the data is received in a plurality of formats, and wherein the data comprises directions for performing a task by the first user under direction from the second user.

[0240] 15. The method of any clause herein, further comprising projecting the data, using the first vocational mask, to a retina of the first user, wherein the data is presented in one or more of the plurality of formats.

[0241] 16. The method of any clause herein, wherein sharing the virtual reality session with the computing device associated with a second user comprises sharing the virtual reality session with a second vocational mask worn by the second user.

[0242] 17. The method of any clause herein, wherein the plurality of formats include one or more of the following:

[0243] image data;

[0244] video data;

[0245] audio data; and

[0246] haptic feedback.

[0247] 18. The method of any clause herein, further comprising sharing the virtual reality session when the first user and the second user being located remotely with respect to one another.

[0248] 19. The method of any clause herein, further comprising sharing the virtual reality session among a plurality of computing devices associated with respective ones of a plurality of users, wherein the plurality of computing devices includes the first vocational mask and one or more additional vocational masks worn by respective ones of a subset of a plurality of users, wherein the subset of the plurality of users includes the first user and one or more additional users, wherein the one or more additional users are apprentices.

[0249] 20. The method of any clause herein, further comprising the first user providing, using the first vocational mask, feedback to the computing device associated with the

second user, wherein the feedback is provided in one or more of the plurality of formats.

[0250] 21. The method of any clause herein, wherein executing the instructions stored in the memory device comprises executing an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

[0251] 22. The method of any clause herein, wherein the task is welding, and wherein the method further comprises:

[0252] monitoring, using the artificial intelligence agent, one or more properties of a weld formed by the first user; and

[0253] adjusting, using the artificial intelligence agent, one or more properties of the weld based on the monitoring.

[0254] 23. A non-transitory computer-readable medium storing instructions that, when executed by a processing device of a first computing system, cause the first computing system to:

[0255] generate, for a first user wearing a first vocational mask that comprises the first computing system, a virtual reality session, wherein the first vocational mask includes a virtual retinal display;

[0256] share the virtual reality session with a second computing system associated with a second user, wherein sharing the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with a second computing system associated with the second user, wherein the first user is an apprentice and wherein the second user is a master; and

[0257] present, to the first user, using the first vocational mask, data received from the second computing system associated with the second user, wherein the data is received in a plurality of formats, and wherein the data comprises directions for performing a task by the first user under direction from the second user.

[0258] 24. The non-transitory computer-readable medium of any clause herein, wherein the first vocational mask includes a virtual retinal display, and wherein the instructions are executable to cause data in at least a subset of the plurality of formats as visual data displayed to the first user via the virtual retinal display.

[0259] 25. The non-transitory computer-readable medium of any clause herein, wherein the subset of the plurality of formats includes video data and image data.

[0260] 26. The non-transitory computer-readable medium of any clause herein, wherein the first vocational mask includes at least one haptic peripheral device, and wherein the instructions are executable to cause haptic feedback, as one of the plurality of formats, to be presented to the first user via the at least one haptic peripheral device.

[0261] 27. The non-transitory computer-readable medium of any clause herein, wherein the instructions, when executed by the processing device of the first computing system, cause the first computing system to execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

[0262] 28. The non-transitory computer-readable medium of any clause herein, wherein the instructions, when

executed by the processing device of the first computing system, cause the first computing system to share the virtual reality session among a plurality of computing devices associated with respective ones of a plurality of users, wherein the plurality of computing devices includes the first vocational mask and one or more additional vocational masks worn by respective ones of a subset of a plurality of users, wherein the subset of the plurality of users includes the first user and one or more additional users, wherein the one or more additional users are apprentices.

[0263] 29. The non-transitory computer-readable medium of any clause herein, wherein the instructions, when executed by the processing device of the first computing system, cause the first computing system to send feedback data to the second computing system.

[0264] 30. A vocational mask configured to be worn by a first user, wherein the vocational mask comprises:

[0265] a virtual retinal display;

[0266] a memory device storing instructions;

[0267] one or more peripheral haptic devices; and

[0268] a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the processing device executes the instructions to:

[0269] share a virtual reality session with a computing device associated with a second user, wherein the vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master;

[0270] receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats from the second user, the data comprising directions for performing a task by the first user under direction from the second user;

[0271] cause the vocational mask to present, via at least one of the one or more peripheral haptic devices, haptic feedback to the first user; and

[0272] execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

[0273] The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described embodiments. However, it should be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of specific embodiments are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the described embodiments to the precise forms disclosed. It should be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

1. A system comprising:

a first vocational mask configured to be worn by a first user, wherein the first vocational mask comprises:

a virtual retinal display;

a memory device having instructions stored thereon; and

a processing device communicatively coupled to the memory device and the virtual retinal display,

wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to:

share a virtual reality session with a computing device associated with a second user, wherein the first vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master; and receive and present data, via the virtual retinal display to the first user, wherein the data is received in selected ones of a plurality of formats from the second user and comprises directions for performing a task by the first user under direction from the second user.

2. The system of claim 1, wherein the first vocational mask is configured to receive the data from a second vocational mask worn by the second user.

3. The system of claim 1, wherein the plurality of formats includes one or more of the following:

audio data;
video data; and
image data.

4. The system of claim 1, wherein the first vocational mask includes one or more peripheral haptic devices, and wherein the instructions, when executed by the processing device, are configured to cause the first vocational mask to present, as one of the plurality of formats, haptic feedback to the first user.

5. The system of claim 1 wherein the first vocational mask is configured to project data in at least one of the plurality of formats to a retina of the first user via the virtual retinal display.

6. The system of claim 1, wherein the first vocational mask is configured to convey feedback data in one or more of the plurality of formats to the computing device of the second user.

7. The system of claim 1, wherein the first user is located remotely with respect to the second user.

8. The system of claim 1, further comprising a plurality of vocational masks including the first vocational mask, wherein each of the plurality of vocational masks is configured to conduct bidirectional communications with the computing device associated with the second user.

9. The system of claim 8, wherein at least a subset of the plurality of vocational masks are configured to receive and present data in a plurality of formats from the computing device associated with the second user, wherein respective users of the subset are apprentices receiving the directions for performing the task.

10. The system of claim 1, wherein the instructions, when executed by the processing device, are configured to execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

11. The system of claim 10, wherein the task is welding, and wherein the artificial intelligence agent is trained to monitor one or more properties of a weld formed by the first user, and further trained to adjust directions to the first user based on the one or more properties of the weld.

12. A method comprising:

wearing, by a first user, a first vocational mask, the first vocational mask having a virtual retinal display, a memory device storing instructions, and a processing device communicatively coupled to the memory device and the virtual retinal display;

executing, using the processing device, the instructions stored in the memory device;

sharing, in response to execution of the instructions, a virtual reality session with a computing device associated with a second user, wherein sharing the virtual reality session comprises the first user conducting bidirectional communications, via the first vocational mask, with the computing device associated with the second user, wherein the first user is an apprentice and the second user is a master; and

presenting, to the first user using the first vocational mask, data received from the computing device associated with the second user, wherein the data is received in a plurality of formats, and wherein the data comprises directions for performing a task by the first user under direction from the second user.

13. The method of claim 12, further comprising projecting the data, using the first vocational mask, to a retina of the first user, wherein the data is presented in one or more of the plurality of formats.

14. The method of claim 12, wherein sharing the virtual reality session with the computing device associated with a second user comprises sharing the virtual reality session with a second vocational mask worn by the second user.

15. The method of claim 12, wherein the plurality of formats include one or more of the following:

image data;
video data;
audio data; and
haptic feedback.

16. The method of claim 12, further comprising sharing the virtual reality session among a plurality of computing devices associated with respective ones of a plurality of users, wherein the plurality of computing devices includes the first vocational mask and one or more additional vocational masks worn by respective ones of a subset of a plurality of users, wherein the subset of the plurality of users includes the first user and one or more additional users, wherein the one or more additional users are apprentices, and wherein at least one of the plurality of users is located remotely with respect to the second user.

17. The method of claim 12, further comprising the first user providing, using the first vocational mask, feedback to the computing device associated with the second user, wherein the feedback is provided in one or more of the plurality of formats.

18. The method of claim 12, wherein executing the instructions stored in the memory device comprises executing an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

19. The method of claim 12, wherein the task is welding, and wherein the method further comprises:

monitoring, using the artificial intelligence agent, one or more properties of a weld formed by the first user; and
adjusting, using the artificial intelligence agent, one or more properties of the weld based on the monitoring.

20. A vocational mask configured to be worn by a first user, wherein the vocational mask comprises:

- a virtual retinal display;
- a memory device storing instructions;
- one or more peripheral haptic devices; and
- a processing device communicatively coupled to the memory device and the virtual retinal display, wherein the processing device executes the instructions to:
 - share a virtual reality session with a computing device associated with a second user, wherein the vocational mask is configured to conduct bidirectional communications with the computing device of the second user, wherein the first user is an apprentice and the second user is a master;
 - receive and present, via the virtual retinal display to the first user, data in selected ones of a plurality of formats from the second user, the data comprising directions for performing a task by the first user under direction from the second user;
 - cause the vocational mask to present, via at least one of the one or more peripheral haptic devices, haptic feedback to the first user; and
 - execute an artificial intelligence agent trained to monitor one or more aspects of the virtual reality session and further trained to provide, in real time, directions for performing the task to the first user during the virtual reality session.

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